

# WEYL-CHAMBER GEOMETRY, POINCARÉ INEQUALITY MEETS SCHATTEN CLASS PROPERTIES OF DUNKL-RIESZ TRANSFORM COMMUTATORS

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ABSTRACT.

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## 1. PRELIMINARIES

**1.1. Harmonic analysis in the Dunkl setting.** Let  $\mathbb{R}^N$  be the Euclidean space equipped with the usual scalar product  $\langle \cdot, \cdot \rangle$  and the induced norm  $\| \cdot \|$ . For a nonzero vector  $v \in \mathbb{R}^N$ , the reflection  $\sigma_v$  with respect to the hyperplane  $v^\perp$  orthogonal to  $v$  is defined by

$$\sigma_v x := x - 2 \frac{\langle x, v \rangle}{\|v\|^2} v.$$

A finite set  $R \subset \mathbb{R}^N \setminus \{0\}$  is called a root system if  $\sigma_v(R) = R$  for every  $v \in R$ . A root system  $R$  is called a normalized reduced root system if  $\|v\|^2 = 2$  for every  $v \in R$ . The finite group  $G$  generated by the reflections  $\{\sigma_v\}_{v \in R}$  is called the Weyl group (reflection group) of the root system  $R$ . Let  $k : R \rightarrow [0, \infty)$  be a multiplicity function, which is a  $G$ -invariant function fixed throughout the paper. We will denote by  $\mathcal{O}(x) = \{\sigma(x) : \sigma \in G\}$  the  $G$ -orbit of a point  $x \in \mathbb{R}^N$ , and  $\mathcal{O}(E) = \bigcup_{x \in E} \mathcal{O}(x)$  the  $G$ -orbit of  $E \subset \mathbb{R}^N$ .

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2010 *Mathematics Subject Classification.* 47B10, 42B20, 43A85.

*Key words and phrases.* Schatten class, Riesz transform commutators, Dunkl operator, Besov space, Nearly weakly orthogonal sequence.

Define the Dunkl measure  $dw$  on  $\mathbb{R}^N$  by

$$dw(x) := \prod_{v \in R} |\langle x, v \rangle|^{k(v)} dx,$$

where  $dx$  stands for the Lebesgue measure on  $\mathbb{R}^N$ . Let  $B(x, r) := \{y \in \mathbb{R}^N : \|x - y\| < r\}$  be the Euclidean ball centred at  $x$  with radius  $r > 0$ . Let  $\mathbf{N} := N + \sum_{v \in R} k(v)$ . Note that

$$w(B(tx, tr)) = t^{\mathbf{N}} w(B(x, r)), \text{ for } x \in \mathbb{R}^N, t, r > 0.$$

Furthermore,

$$(1) \quad w(B(x, r)) \sim r^{\mathbf{N}} \prod_{v \in R} (|\langle x, v \rangle| + r)^{k(v)},$$

where the implicit constant only depends on  $N, R$  and  $k$ . It follows from (1) that there exists a constant  $C \geq 1$  such that for every  $x \in \mathbb{R}^N$ ,

$$(2) \quad C^{-1} \left( \frac{r_2}{r_1} \right)^{\mathbf{N}} \leq \frac{w(B(x, r_2))}{w(B(x, r_1))} \leq C \left( \frac{r_2}{r_1} \right)^{\mathbf{N}}, \text{ for } 0 < r_1 < r_2,$$

so the numbers  $\mathbf{N}$  and  $N$  are called the upper and lower homogeneous dimensions, respectively.

[Add the definition of  $A_\infty$ . Zhijie will do it.]

**Lemma 1.1.** *Let*

$$dw(x) = \varpi(x) dx, \quad \varpi(x) = \prod_{v \in R_+} |\langle x, v \rangle|^{2k(v)}.$$

Then  $\varpi \in A_\infty(\mathbb{R}^N, dx)$ .

*Proof.* The assertion is immediate if  $k \equiv 0$ . Otherwise, choose  $p > 1$  sufficiently large so that

$$\frac{2|R_+|k(v)}{p-1} < 1, \quad v \in R_+.$$

For every  $0 < \beta < 1$ ,  $v \in R_+$ , and Euclidean ball  $B(x_0, r)$ , we apply an orthogonal change of variables to deduce that

$$\begin{aligned} \frac{1}{|B(x_0, r)|} \int_{B(x_0, r)} |\langle x, v \rangle|^{-\beta} dx &\lesssim \frac{1}{r|v|} \int_{\langle x_0, v \rangle - |v|r}^{\langle x_0, v \rangle + |v|r} |s|^{-\beta} ds \\ &= (|v|r)^{-\beta} \int_{\frac{\langle x_0, v \rangle}{|v|r} - 1}^{\frac{\langle x_0, v \rangle}{|v|r} + 1} |s|^{-\beta} ds \\ &\lesssim (|v|r)^{-\beta} \left( 1 + \frac{|\langle x_0, v \rangle|}{|v|r} \right)^{-\beta} \\ &\lesssim (|\langle x_0, v \rangle| + r)^{-\beta}. \end{aligned}$$

Hence, by Hölder's inequality,

$$\begin{aligned} \frac{1}{|B(x_0, r)|} \int_{B(x_0, r)} \varpi^{-\frac{1}{p-1}} dx &= \frac{1}{|B(x_0, r)|} \int_{B(x_0, r)} \prod_{v \in R_+} |\langle x, v \rangle|^{-\frac{2k(v)}{p-1}} dx \\ &\leq \prod_{v \in R_+} \left( \frac{1}{|B(x_0, r)|} \int_{B(x_0, r)} |\langle x, v \rangle|^{-\frac{2|R_+|k(v)}{p-1}} dx \right)^{\frac{1}{|R_+|}} \\ &\lesssim \prod_{v \in R_+} (|\langle x_0, v \rangle| + r)^{-\frac{2k(v)}{p-1}}. \end{aligned}$$

On the other hand, by (1) and the  $G$ -invariance of  $k$ ,

$$\frac{1}{|B(x_0, r)|} \int_{B(x_0, r)} \varpi dx = \frac{w(B(x_0, r))}{|B(x_0, r)|} \lesssim \prod_{v \in R_+} (|\langle x_0, v \rangle| + r)^{2k(v)}.$$

Therefore,

$$\left( \frac{1}{|B|} \int_B \varpi dx \right) \left( \frac{1}{|B|} \int_B \varpi^{-\frac{1}{p-1}} dx \right)^{p-1} \lesssim 1$$

uniformly over all Euclidean balls  $B$ . Thus  $\varpi \in A_p(\mathbb{R}^N, dx)$ , and hence  $\varpi \in A_\infty(\mathbb{R}^N, dx)$ .  $\square$

The Dunkl operators  $T_\xi$  are defined by

$$T_\xi f(x) := \partial_\xi f(x) + \sum_{v \in R} \frac{k(v)}{2} \langle v, \xi \rangle \frac{f(x) - f(\sigma_v(x))}{\langle v, x \rangle},$$

where  $\partial_\xi f$  denotes the directional derivative of  $f$  in the direction  $\xi$ . Furthermore, the Dunkl Laplacian is defined by

$$\Delta_{\text{Dunkl}} f(x) := \sum_{j=1}^N T_{e_j}^2 f(x) = \Delta_{\mathbb{R}^N} f(x) + \sum_{v \in R} k(v) \delta_v f(x),$$

where  $\Delta_{\mathbb{R}^N}$  is the Laplacian on  $\mathbb{R}^N$  and  $\delta_v$  is defined by

$$\delta_v f(x) := \frac{\partial_v f(x)}{\langle v, x \rangle} - \frac{f(x) - f(\sigma_v(x))}{\langle v, x \rangle^2}.$$

In this article, we consider the Dunkl-Riesz transforms defined by  $R_{\text{Dunkl}, j} := T_{e_j}(-\Delta_{\text{Dunkl}})^{-1/2}$ ,  $j = 1, 2, \dots, N$ .

**Definition 1.2.** The Weyl chambers are defined as the closures of connected components of the set

$$(3) \quad \{x \in \mathbb{R}^N : \langle x, v \rangle \neq 0 \text{ for all } v \in R\}.$$

Define  $d(x, y) := \min_{\sigma \in G} \|x - \sigma(y)\|$  as the distance between two  $G$ -orbits  $\mathcal{O}(x)$  and  $\mathcal{O}(y)$ . Recall from [7, Chapter VII, proof of Theorem 2.12] that

$$(4) \quad d(x, y) = \|x - y\| \text{ if and only if } x, y \in \Gamma$$

for some closed Weyl chamber  $\Gamma$ .

It is clear that  $\mathcal{O}(B(x, r)) = \{y \in \mathbb{R}^N : d(x, y) < r\}$ . Moreover,

$$w(B(x, r)) \leq w(\mathcal{O}(B(x, r))) \leq |G|w(B(x, r))$$

for all  $x \in \mathbb{R}^N$  and  $r > 0$ . We also denote  $B^d(x, r) := \{y \in \mathbb{R}^N : d(x, y) < r\}$ , i.e.,  $B^d(x, r) = \mathcal{O}(B(x, r))$ .

Denote by  $K_j(x, y)$  the integral kernel of  $R_{\text{Dunkl}, j}$ . For every  $x, y \in \mathbb{R}^N$  and  $r > 0$ , let

$$V(x, y, r) := \max\{w(B(x, r)), w(B(y, r))\}.$$

The following lemma establishes derivative estimates for the Dunkl–Riesz transform kernel.

**Lemma 1.3.** *Assume that  $N \geq 1$ . Then for every pair of multi-indices  $\alpha, \beta$ , every  $j \in \{1, \dots, N\}$ , and every  $x, y \in \mathbb{R}^N$  with  $d(x, y) > 0$ ,*

$$(5) \quad \left| \partial_x^\alpha \partial_y^\beta K_{\text{Dunkl}, j}(x, y) \right| \lesssim_{\alpha, \beta, R, k} \frac{d(x, y)}{\|x - y\|} \frac{d(x, y)^{-(|\alpha| + |\beta|)}}{V(x, y, d(x, y))}.$$

*Proof. Case 1.* Suppose that  $|\alpha| + |\beta| = 0$ . By [5, Theorem 1.1],

$$|K_{\text{Dunkl}, j}(x, y)| \lesssim \frac{d(x, y)}{\|x - y\|} \frac{1}{w(B(x, d(x, y)))}.$$

By the definition of  $d$ , the  $G$ -invariance of  $dw$ , and (2),

$$V(x, y, d(x, y)) \lesssim w(B(x, d(x, y))),$$

and hence (5) follows.

**Case 2.** Suppose that  $|\alpha| + |\beta| \geq 1$ . By [?, Theorem 3.1],

$$\left| \partial_x^\alpha \partial_y^\beta h_t(x, y) \right| \lesssim_{\alpha, \beta, R, k} t^{-\frac{|\alpha| + |\beta|}{2}} \left( 1 + \frac{\|x - y\|}{\sqrt{t}} \right)^{-2} V(x, y, \sqrt{t})^{-1} e^{-cd(x, y)^2/t}.$$

Recall from the proof of Lemma ?? that

$$K_{\text{Dunkl}, j}(x, y) = -\frac{C}{2} \int_0^\infty (y_j - x_j) h_t(x, y) \frac{dt}{t^{3/2}}.$$

Since

$$\partial_x^\alpha \partial_y^\beta ((y_j - x_j) h_t(x, y)) = (y_j - x_j) \partial_x^\alpha \partial_y^\beta h_t(x, y) - \alpha_j \partial_x^{\alpha - e_j} \partial_y^\beta h_t(x, y) + \beta_j \partial_x^\alpha \partial_y^{\beta - e_j} h_t(x, y),$$

where the corresponding term is omitted if  $\alpha_j = 0$  or  $\beta_j = 0$ , we obtain

$$\begin{aligned} \left| \partial_x^\alpha \partial_y^\beta K_{\text{Dunkl}, j}(x, y) \right| &\lesssim \frac{1}{\|x - y\|} \int_0^\infty t^{-\frac{|\alpha| + |\beta| + 1}{2}} V(x, y, \sqrt{t})^{-1} e^{-cd(x, y)^2/t} dt \\ &\quad + \frac{1}{\|x - y\|^2} \int_0^\infty t^{-\frac{|\alpha| + |\beta|}{2}} V(x, y, \sqrt{t})^{-1} e^{-cd(x, y)^2/t} dt. \end{aligned}$$

By (2),

$$V(x, y, \sqrt{t})^{-1} \lesssim V(x, y, d(x, y))^{-1} \begin{cases} \left( \frac{d(x, y)}{\sqrt{t}} \right)^N, & 0 < t \leq d(x, y)^2, \\ \left( \frac{d(x, y)}{\sqrt{t}} \right)^N, & t \geq d(x, y)^2. \end{cases}$$

Hence,

$$\begin{aligned} &\int_0^\infty t^{-\frac{|\alpha| + |\beta| + 1}{2}} V(x, y, \sqrt{t})^{-1} e^{-cd(x, y)^2/t} dt \\ &\lesssim \frac{1}{V(x, y, d(x, y))} \left( d(x, y)^N \int_0^{d(x, y)^2} t^{-\frac{N + |\alpha| + |\beta| + 1}{2}} e^{-cd(x, y)^2/t} dt + d(x, y)^N \int_{d(x, y)^2}^\infty t^{-\frac{N + |\alpha| + |\beta| + 1}{2}} dt \right) \\ &\lesssim \frac{d(x, y)^{1 - |\alpha| - |\beta|}}{V(x, y, d(x, y))}. \end{aligned}$$

The first integral in the parentheses is estimated by the change of variables  $s = d(x, y)^2/t$ , while the second one converges since  $N + |\alpha| + |\beta| > 1$ .

**Case 2.1.** Suppose that  $N + |\alpha| + |\beta| > 2$ . We apply the same argument to see that

$$\begin{aligned} & \int_0^\infty t^{-\frac{|\alpha|+|\beta|}{2}} V(x, y, \sqrt{t})^{-1} e^{-cd(x, y)^2/t} dt \\ & \lesssim \frac{1}{V(x, y, d(x, y))} \left( d(x, y)^N \int_0^{d(x, y)^2} t^{-\frac{N+|\alpha|+|\beta|}{2}} e^{-cd(x, y)^2/t} dt + d(x, y)^N \int_{d(x, y)^2}^\infty t^{-\frac{N+|\alpha|+|\beta|}{2}} dt \right) \\ & \lesssim \frac{d(x, y)^{2-|\alpha|-|\beta|}}{V(x, y, d(x, y))}. \end{aligned}$$

Since  $d(x, y) \leq \|x - y\|$ , it follows that

$$|\partial_x^\alpha \partial_y^\beta K_{\text{Dunkl}, j}(x, y)| \lesssim_{\alpha, \beta, R, k} \frac{d(x, y)}{\|x - y\|} \frac{d(x, y)^{-(|\alpha|+|\beta|)}}{V(x, y, d(x, y))}.$$

**Case 2.2.** Suppose that  $N = 1$  and  $|\alpha| + |\beta| = 1$ . For the terms in which a derivative falls on  $y_j - x_j$ , retaining the factor  $(1 + \|x - y\|/\sqrt{t})^{-2}$  and splitting the integral at  $d(x, y)^2$  and  $\|x - y\|^2$ , we obtain from (2)

$$\begin{aligned} & \int_0^\infty t^{-3/2} \left( 1 + \frac{\|x - y\|}{\sqrt{t}} \right)^{-2} V(x, y, \sqrt{t})^{-1} e^{-cd(x, y)^2/t} dt \\ & \lesssim \frac{d(x, y)^N}{\|x - y\|^2 V(x, y, d(x, y))} \int_0^{d(x, y)^2} t^{-\frac{N+1}{2}} e^{-cd(x, y)^2/t} dt \\ & \quad + \frac{d(x, y)}{\|x - y\|^2 V(x, y, d(x, y))} \int_{d(x, y)^2}^{\|x - y\|^2} \frac{dt}{t} + \frac{d(x, y)}{V(x, y, d(x, y))} \int_{\|x - y\|^2}^\infty \frac{dt}{t^2} \\ & \lesssim \frac{1}{\|x - y\| V(x, y, d(x, y))}, \end{aligned}$$

where we used

$$\frac{d(x, y)}{\|x - y\|} \log \frac{\|x - y\|}{d(x, y)} \lesssim 1.$$

Combining this estimate with the preceding estimate for the first term, we obtain

$$|\partial_x^\alpha \partial_y^\beta K_{\text{Dunkl}, j}(x, y)| \lesssim_{\alpha, \beta, R, k} \frac{1}{\|x - y\| V(x, y, d(x, y))} = \frac{d(x, y)}{\|x - y\|} \frac{d(x, y)^{-1}}{V(x, y, d(x, y))}.$$

This completes the proof of Lemma 1.3.  $\square$

## 1.2. Systems of dyadic cubes.

**Notation 1.4.** For every dyadic cube  $Q \in \mathbb{D}$ , we denote by  $c_Q$  and  $\ell(Q)$  its center and side-length, respectively.

**Notation 1.5.** Let

$$\mathbb{D}^0 := \bigcup_{k \in \mathbb{Z}} \mathbb{D}_k^0$$

be the standard system of dyadic cubes over  $\mathbb{R}^N$ . Here each dyadic cube is a product of half-open intervals

$$[m_1 2^{-k}, (m_1 + 1) 2^{-k}) \times \cdots \times [m_N 2^{-k}, (m_N + 1) 2^{-k}),$$

with  $(m_1, \dots, m_N) \in \mathbb{Z}^N$ . Moreover, we let

$$\mathbb{D}_\Gamma := \bigcup_{k \in \mathbb{Z}} \mathbb{D}_{\Gamma, k}$$

be the system of dyadic cubes over a fixed chamber  $\Gamma$ , whose construction will be given in Corollary 2.7.

**1.3. Nearly weakly orthonormal sequences.** Rochberg and Semmes [12] introduced the notion of *nearly weakly orthonormal* (NWO) sequences of functions. This notion has since proved highly effective in characterizing Schatten class membership of singular integral commutators in a variety of contexts (see [3, 8, 11, 1, 2, 9, 12, 14, 13]). The properties of NWO sequences were originally investigated in the Euclidean setting (equipped with Lebesgue measure), but the extension to spaces of homogeneous type is straightforward once a system of dyadic cubes is available in this general framework (see [8, Section 13] for a complete proof). For our purposes, we adapt the extension to the Dunkl setting  $(\mathbb{R}^N, \|\cdot\|, w)$  rather than recalling the general definition of NWO sequences and the general extension to spaces of homogeneous type.

**Definition 1.6.** Let  $\Gamma$  be a chamber. Let  $\Phi$  be as in Corollary 2.6 and let  $\mathbb{D}_\Gamma$  be as in Corollary 2.7. Let  $c > 0$  and  $2 < r \leq \infty$ . A sequence of functions  $\{e_I\}_{I \in \mathbb{D}_\Gamma}$  is called  $(r, c)$ -NWO if

- (1)  $\text{supp}(e_I) \subseteq cI$  for every  $I \in \mathbb{D}_\Gamma$ ;
- (2)  $\|e_I\|_{L_r(\mathbb{R}^N, w(x)dx)} \leq w(I)^{\frac{1}{r} - \frac{1}{2}}$  for every  $I \in \mathbb{D}_\Gamma$ ;

**Lemma 1.7.** Let  $0 < p < \infty$ . Let  $\Gamma$  be a chamber. Let  $\Phi$  be as in Corollary 2.6 and let  $\mathbb{D}_\Gamma$  be as in Corollary 2.7. Suppose that  $A$  is an integral operator on  $L_2(\Gamma, w(x)dx)$  with integral kernel

$$(x, y) \rightarrow \sum_{I \in \mathbb{D}_\Gamma} \lambda_I e_I(x) f_I(y), \quad x, y \in \Gamma,$$

where  $\{e_I\}_{I \in \mathbb{D}_\Gamma}$  and  $\{f_I\}_{I \in \mathbb{D}_\Gamma}$  are  $(r_1, c_1)$ -NWO and  $(r_2, c_2)$ -NWO sequences, respectively. We have

$$\begin{aligned} \|A\|_{\mathcal{L}_{p, \infty}(L_2(\Gamma, w(x)dx))} &\leq C_{\Phi, \kappa, r_1, c_1, r_2, c_2, p} \|\{\lambda_I\}_{I \in \mathbb{D}_\Gamma}\|_{p, \infty}, \\ \text{dist}_{\mathcal{L}_{p, \infty}(L_2(\Gamma, w(x)dx))} \left( A, (\mathcal{L}_{p, \infty})_0(L_2(\Gamma, w(x)dx)) \right) &\leq \\ &\leq C_{\Phi, \kappa, r_1, c_1, r_2, c_2, p} \text{dist}_{l_{p, \infty}(\mathbb{D}_\Gamma)} \left( \{\lambda_I\}_{I \in \mathbb{D}_\Gamma}, (l_{p, \infty})_0(\mathbb{D}_\Gamma) \right). \end{aligned}$$

*Proof.* state a similar lemma with  $\Gamma$  replaced by  $\mathbb{R}_+^{n_1} \times \mathbb{R}^{n_2}$  and more-or-less-generic weight instead of  $w$ . Provide reference to the LATTER assertion. this assertion would trivially follow from the to-be-stated one. This property can be found in [12, (1.9)] (see also [8, Section 7]).  $\square$

## 2. ANALYTIC TOOLS FOR WEYL CHAMBER GEOMETRY

**Convention.** Throughout this paper, we will use the following notation:

- Let  $R \subset \mathbb{R}^N$  be a finite normalized reduced root system;
- Let  $W := \text{span}(R)$  with  $N_0 := \dim W$ ;
- Let  $\Omega := \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0}$ ;

- Let  $\Gamma_+$  be a fixed open Weyl chamber (see Definition 2.1) and  $\Gamma$  be its closure.

**Definition 2.1.** [6, Definition 2.7] *A connected component of*

$$(1) \quad \{x \in \mathbb{R}^N : \langle x, v \rangle \neq 0 \text{ for all } v \in R\}$$

*is called an open Weyl chamber. Its closure is called a Weyl chamber. Define the corresponding sets of positive and negative roots by*

$$R_+ := \{\alpha \in R : \langle x, \alpha \rangle > 0 \text{ for all } x \in \Gamma_+\}, \quad R_- := -R_+.$$

**Definition 2.2.** [6, Definition 2.9] *A root  $\alpha \in R_+$  is called simple in  $R_+$  if it cannot be written as*

$$\alpha = \lambda_1 \alpha_1 + \lambda_2 \alpha_2 \quad \text{with } \lambda_1, \lambda_2 \geq 1 \text{ and } \alpha_1, \alpha_2 \in R_+.$$

*The set  $\Delta \subset R_+$  of all simple roots is called the simple system.*

**Lemma 2.3.** *With the notation given in Definition 2.2,  $\Delta$  is a basis of  $W$ . If we denote by  $\Delta = \{\alpha_1, \alpha_2, \dots, \alpha_{N_0}\}$ , then the open and closed chambers can be written as*

$$(2) \quad \Gamma_+ = \left\{x \in \mathbb{R}^N : \langle x, \alpha_i \rangle > 0, i = 1, \dots, N_0\right\},$$

$$(3) \quad \Gamma = \left\{x \in \mathbb{R}^N : \langle x, \alpha_i \rangle \geq 0, i = 1, \dots, N_0\right\},$$

*and every root  $\alpha \in R$  admits a unique expansion*

$$\alpha = \sum_{i=1}^{N_0} c_i \alpha_i$$

*such that either  $c_i \geq 0$  for all  $i$  (when  $\alpha \in R_+$ ), or  $c_i \leq 0$  for all  $i$  (when  $\alpha \in R_-$ ). In particular, the simple system  $\Delta$  is uniquely determined by  $\Gamma_+$  (equivalently, by  $\Gamma$ ).*

*Proof.* Let  $\Delta = \{\alpha_1, \dots, \alpha_n\} \subset R_+$  be the set of simple roots in  $R_+$ . By [6, Proposition 2.10], every  $\alpha \in R_+$  is a nonnegative linear combination of  $\alpha_1, \dots, \alpha_n$ . Since  $R = R_+ \cup R_-$  (this follows from the fact that for each  $\alpha \in R$ , the continuous function  $x \mapsto \langle x, \alpha \rangle$  does not vanish on the connected set  $\Gamma_+$ ) and  $R_- = -R_+$ , it follows that  $R \subset \text{span}(\Delta)$  and hence,

$$W = \text{span}(R) \subset \text{span}(\Delta).$$

The reverse inclusion  $\text{span}(\Delta) \subset W$  is immediate from  $\Delta \subset R \subset W$ , so  $\text{span}(\Delta) = W$ . On the other hand, it follows from [6, Theorem 2.16] that the simple roots are linearly independent. Consequently,  $\Delta$  is a basis of  $W$ , and therefore  $n = \dim W = N_0$ . Moreover, every  $\alpha \in R$  has a unique expansion  $\alpha = \sum_{i=1}^{N_0} c_i \alpha_i$ . If  $\alpha \in R_+$ , the coefficients can be chosen nonnegative by [6, Proposition 2.10]. If  $\alpha \in R_- = -R_+$ , then  $-\alpha \in R_+$  has an expansion with all coefficients nonnegative, and multiplying by  $-1$  gives an expansion of  $\alpha$  with all coefficients nonpositive. The descriptions (2) and (3) follow from [6, Corollary 2.11].

To see that  $\Delta$  is uniquely determined by  $\Gamma_+$ , note that  $\Gamma_+$  determines  $R_+$  by Definition 2.1, and the set of simple roots in  $R_+$  is characterized intrinsically by Definition 2.2. Finally, since  $\Gamma_+$  is the (unique) interior component of  $\Gamma$ , specifying  $\Gamma$  determines  $\Gamma_+$  and vice versa. This completes the proof of Lemma 2.3.  $\square$

**Notation 2.4.** With the notation given in Lemma 2.3, denote by

$$\Delta = \{\alpha_1, \dots, \alpha_{N_0}\}$$

the simple system associated with  $\Gamma$ .

**Lemma 2.5.** Each Weyl chamber  $\Gamma$  admits the decomposition

$$\Gamma = K(\Gamma) \times W^\perp, \quad K(\Gamma) := \{w \in W : \langle w, \alpha \rangle \geq 0 \text{ for all } \alpha \in \Delta\}.$$

*Proof.* Write  $x = w + z$  with  $w \in W$  and  $z \in W^\perp$ . Since  $\alpha_i \in W$ , we have  $\langle x, \alpha_i \rangle = \langle w, \alpha_i \rangle$  for all  $i = 1, \dots, N_0$ . This, together with (3), implies that  $x \in \Gamma$  if and only if  $w \in K(\Gamma)$ . This implies that  $\Gamma = K(\Gamma) \times W^\perp$ , and thus completes the proof.  $\square$

**Corollary 2.6.** Let  $\Phi_{W^\perp}$  be any linear isomorphism from  $W^\perp$  to  $\mathbb{R}^{N-N_0}$ , and let

$$\Phi_W : W \longrightarrow \mathbb{R}^{N_0}, \quad \Phi_W(w) := (\langle w, \alpha_1 \rangle, \dots, \langle w, \alpha_{N_0} \rangle),$$

$$\Phi : \mathbb{R}^N = W \oplus W^\perp \longrightarrow \mathbb{R}^{N_0} \oplus \mathbb{R}^{N-N_0} \cong \mathbb{R}^N, \quad \Phi(w + z) := (\Phi_W(w), \Phi_{W^\perp}(z)).$$

Then  $\Phi$  is a linear isomorphism and

$$\Phi(\Gamma) = \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0}.$$

*Proof.* By Lemma 2.5 we have  $\Gamma = K(\Gamma) \times W^\perp$ , where

$$K(\Gamma) = \{w \in W : \langle w, \alpha_i \rangle \geq 0, i = 1, \dots, N_0\}.$$

Since  $\Delta$  is a basis of  $W$ , the Gram matrix  $G = (\langle \alpha_i, \alpha_j \rangle)_{i,j=1}^{N_0}$  is nonsingular. Hence,  $\Phi_W : W \longrightarrow \mathbb{R}^{N_0}$  is a linear isomorphism. Moreover, for  $w \in W$  we have  $w \in K(\Gamma)$  if and only if  $\Phi_W(w) \in \mathbb{R}_+^{N_0}$ , so

$$\Phi_W(K(\Gamma)) = \mathbb{R}_+^{N_0}.$$

By construction,  $\Phi$  is a linear isomorphism (as a direct sum of isomorphisms), and therefore

$$\Phi(\Gamma) = \Phi(K(\Gamma) \times W^\perp) = \Phi_W(K(\Gamma)) \times \Phi_{W^\perp}(W^\perp) = \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0},$$

which proves the desired conclusion.  $\square$

**Corollary 2.7.** Let  $\Gamma$  be an arbitrary Weyl chamber. There exists a system of dyadic cubes over  $\Gamma$ .

*Proof.* Let  $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$  be the linear isomorphism from Corollary 2.6 so that

$$\Phi(\Gamma) = \Omega := \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0}.$$

For each  $k \in \mathbb{Z}$ , define  $\mathbb{D}_{\Omega,k}^0$  to be the family of half-open cubes

$$Q = \prod_{i=1}^N I_i, \quad I_i = [m_i 2^{-k}, (m_i + 1) 2^{-k}),$$

where  $m_i \in \mathbb{Z}_+$  for  $1 \leq i \leq N_0$  and  $m_i \in \mathbb{Z}$  for  $N_0 < i \leq N$ . Set

$$\mathbb{D}_\Omega^0 := \bigcup_{k \in \mathbb{Z}} \mathbb{D}_{\Omega,k}^0.$$

It is standard that  $\mathbb{D}_\Omega^0$  is a system of dyadic cubes over  $\Omega$ .

Next, for each  $k \in \mathbb{Z}$ , define

$$\mathbb{D}_{\Gamma,k} := \{\Phi^{-1}(Q) : Q \in \mathbb{D}_{\Omega,k}^0\}, \quad \mathbb{D}_{\Gamma} := \bigcup_{k \in \mathbb{Z}} \mathbb{D}_{\Gamma,k}.$$

Since  $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$  is a linear isomorphism, we see that  $\mathbb{D}_{\Gamma}$  is a system of dyadic cubes over  $\Gamma$ .  $\square$

**Notation 2.8.** Let  $\Phi$  be the linear isomorphism given in Corollary 2.6. For every Borel set  $E \subset \Omega$ , define

$$(4) \quad \mu(E) := w(\Phi^{-1}(E)).$$

**Lemma 2.9.** We have

$$(5) \quad d\mu(u, z) = \phi(u) du dz, \quad \phi(u) = |\det(\Phi)|^{-1} \prod_{v \in R_+} \left( \sum_{i=1}^{N_0} a_{v,i} u_i \right)^{2k(v)}.$$

where for each  $v \in R_+$ , the coefficients  $a_{v,i} \geq 0$  are uniquely determined by the simple-root expansion  $v = \sum_{i=1}^{N_0} a_{v,i} \alpha_i$  and satisfy  $(a_{v,1}, \dots, a_{v,N_0}) \neq \mathbf{0}$ .

*Proof.* Since  $\Phi(\Gamma) = \Omega$ , we have  $\Phi^{-1}(\Omega) = \Gamma$ . Hence, for every nonnegative Borel function  $f$  on  $\Omega$ ,

$$\int_{\Omega} f(u, z) d\mu(u, z) = \int_{\Gamma} f(\Phi(x)) dw(x).$$

We make the linear change of variables  $(u, z) = \Phi(x)$ , so that  $x = \Phi^{-1}(u, z)$ . Since  $\Phi$  is linear and invertible, the Jacobian is constant and

$$dx = |\det(\Phi)|^{-1} du dz.$$

It follows from the definition of  $\Phi$  that  $u_i = \langle x, \alpha_i \rangle$  for  $i = 1, \dots, N_0$ . This implies that

$$(6) \quad \langle x, v \rangle = \left\langle x, \sum_{i=1}^{N_0} a_{v,i} \alpha_i \right\rangle = \sum_{i=1}^{N_0} a_{v,i} \langle x, \alpha_i \rangle = \sum_{i=1}^{N_0} a_{v,i} u_i.$$

Substituting these identities into the previous integral, we obtain

$$\begin{aligned} \int_{\Omega} f(u, z) d\mu(u, z) &= \int_{\Gamma} f(\Phi(x)) \left( \prod_{v \in R_+} |\langle x, v \rangle|^{2k(v)} \right) dx \\ &= \int_{\Omega} f(u, z) |\det(\Phi)|^{-1} \prod_{v \in R_+} \left( \sum_{i=1}^{N_0} a_{v,i} u_i \right)^{2k(v)} du dz. \end{aligned}$$

Therefore, (5) holds. This completes the proof of Lemma 2.9.  $\square$

**Lemma 2.10.** Let  $\Gamma_1$  and  $\Gamma_2$  be two Weyl chambers. Then there exist an integer  $M \geq 0$  and  $\alpha_1, \dots, \alpha_M \in R$  such that, if we set  $\Gamma^{(0)} := \Gamma_2$  and

$$\Gamma^{(k)} := \sigma_{\alpha_k}(\Gamma^{(k-1)}), \quad k = 1, \dots, M,$$

then the following hold:

- (1)  $\alpha_k \in \Delta(\Gamma^{(k-1)})$  for each  $k = 1, \dots, M$ ;
- (2)  $\Gamma^{(M)} = \Gamma_1$ .

*Proof.* If  $\Gamma_1 = \Gamma_2$ , we take  $M = 0$  and there is nothing to prove. Assume  $\Gamma_1 \neq \Gamma_2$ . Since the Weyl group  $G$  acts transitively on the set of Weyl chambers (see [6, Theorem 2.12 (2)]), there exists  $g \in G$  such that  $\Gamma_1 = g(\Gamma_2)$ .

Let  $\Delta(\Gamma_2) = \{\beta_1, \beta_2, \dots, \beta_{N_0}\}$  be the simple system associated with  $\Gamma_2$ . By [6, Theorem 2.12(4)], the Weyl group  $G$  is generated by the simple reflections  $\sigma_{\beta_1}, \sigma_{\beta_2}, \dots, \sigma_{\beta_{N_0}}$ . Hence, there exist indices  $i_1, \dots, i_M \in \{1, \dots, N_0\}$  such that

$$g = \sigma_{\beta_{i_1}} \sigma_{\beta_{i_2}} \cdots \sigma_{\beta_{i_M}}.$$

Define

$$w_0 := \text{Id}, \quad w_k := \sigma_{\beta_{i_1}} \cdots \sigma_{\beta_{i_k}} \quad (k = 1, \dots, M),$$

and set

$$\Gamma^{(k)} := w_k(\Gamma_2) \quad (k = 0, 1, \dots, M).$$

It follows that  $\Gamma^{(0)} = \Gamma_2$  and  $\Gamma^{(M)} = w_M(\Gamma_2) = g(\Gamma_2) = \Gamma_1$ , which implies the statement (2).

For each  $k = 1, \dots, M$ , define

$$\alpha_k := w_{k-1}(\beta_{i_k}) \in R.$$

The rest of the proof is divided into two steps.

• **Step 1.** Verify that  $\Gamma^{(k)} = \sigma_{\alpha_k}(\Gamma^{(k-1)})$ .

Since each  $w \in G$  is orthogonal, for any root  $\beta \in R$  one has the conjugation identity

$$w \sigma_{\beta} w^{-1} = \sigma_{w(\beta)}, \quad w \in G.$$

Applying this with  $w = w_{k-1}$  and  $\beta = \beta_{i_k}$  yields

$$\sigma_{\alpha_k} = \sigma_{w_{k-1}(\beta_{i_k})} = w_{k-1} \sigma_{\beta_{i_k}} w_{k-1}^{-1}.$$

Therefore,

$$\sigma_{\alpha_k}(\Gamma^{(k-1)}) = w_{k-1} \sigma_{\beta_{i_k}} w_{k-1}^{-1}(w_{k-1}(\Gamma_2)) = w_{k-1} \sigma_{\beta_{i_k}}(\Gamma_2) = w_k(\Gamma_2) = \Gamma^{(k)}.$$

• **Step 2.** Verify that  $\alpha_k \in \Delta(\Gamma^{(k-1)})$ .

Recall that if  $\Gamma$  is a Weyl chamber associated with simple system  $\Delta(\Gamma)$ , then

$$\Gamma = \{x \in \mathbb{R}^N : \langle x, \gamma \rangle \geq 0 \text{ for all } \gamma \in \Delta(\Gamma)\}.$$

Applying  $w_{k-1} \in G$  to the above description for  $\Gamma_2$  and using  $\langle w_{k-1}x, \beta \rangle = \langle x, w_{k-1}^{-1}\beta \rangle$ , we obtain

$$\Gamma^{(k-1)} = w_{k-1}(\Gamma_2) = \left\{ x \in \mathbb{R}^N : \langle x, w_{k-1}(\beta_j) \rangle \geq 0 \text{ for } j = 1, \dots, N_0 \right\}.$$

By the uniqueness of the simple system associated with a Weyl chamber (see Lemma 2.3), we conclude that

$$\Delta(\Gamma^{(k-1)}) = \Delta(w_{k-1}(\Gamma_2)) = w_{k-1}(\Delta(\Gamma_2)) = \{w_{k-1}(\beta_1), \dots, w_{k-1}(\beta_{N_0})\}.$$

In particular,  $\alpha_k = w_{k-1}(\beta_{i_k}) \in \Delta(\Gamma^{(k-1)})$ , proving the first statement (1).

This completes the proof of Lemma 2.10.  $\square$

**Definition 2.11.** We say that  $R' \subset \Omega$  is an axis-parallel rectangle if

$$R' := I_1 \times \cdots \times I_N,$$

with  $I_i = [I_i^-, I_i^+] \subset [0, \infty)$  for  $1 \leq i \leq N_0$  and  $I_i = [I_i^-, I_i^+] \subset \mathbb{R}$  for  $i > N_0$ .

Denote by

$$B_\Omega(x_0, r) := \{x \in \Omega : \|x - x_0\| < r\}.$$

The following lemma demonstrates that  $(\Omega, \|\cdot\|, d\mu)$  is a doubling metric measure space.

**Lemma 2.12.** *There exists a constant  $C \geq 1$  such that for every  $x_0 = (u_0, z_0) \in \Omega$  and every  $r > 0$ ,*

$$(7) \quad C^{-1} r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} \leq \mu(B_\Omega(x_0, r)) \leq C r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)},$$

where

$$L_v(u) := \sum_{i=1}^{N_0} a_{v,i} u_i, \quad L_{v,0} := L_v(u_0) = \sum_{i=1}^{N_0} a_{v,i} (u_0)_i.$$

In particular, there exists a constant  $C_{\text{dbl}, \Omega} \geq 1$  such that for every  $x \in \Omega$  and every  $r > 0$ ,

$$\mu(B_\Omega(x, 2r)) \leq C_{\text{dbl}, \Omega} \mu(B_\Omega(x, r)).$$

*Proof.* Fix  $x_0 = (u_0, z_0) \in \Omega$  and  $r > 0$ . For each  $v \in R_+$  define the linear form

$$L_v(u) := a_v \cdot u, \quad a_v := (a_{v,1}, \dots, a_{v,N_0}) \in \mathbb{R}_+^{N_0} \setminus \{0\}.$$

Since  $R_+$  is finite and each  $a_v \neq 0$ , we have

$$L_* := \max_{v \in R_+} \|a_v\| < \infty, \quad m_* := \min_{v \in R_+} \sum_{i=1}^{N_0} a_{v,i} > 0.$$

**Upper bound.** Let  $x = (u, z) \in B_\Omega(x_0, r)$ . Then

$$\|u - u_0\| \leq \|x - x_0\| \leq r.$$

Hence, for every  $v \in R_+$ ,

$$|L_v(u) - L_{v,0}| = |a_v \cdot (u - u_0)| \leq \|a_v\| \cdot \|u - u_0\| \leq L_* r.$$

Since  $L_v(u) \geq 0$  on  $\Omega$  (all coefficients and coordinates are nonnegative), we obtain

$$L_v(u) \leq L_{v,0} + L_* r \leq C_1 (L_{v,0} + r), \quad C_1 := \max\{1, L_*\}.$$

Therefore,

$$\phi(u) = \prod_{v \in R_+} L_v(u)^{2k(v)} \leq C_1^{2 \sum_{v \in R_+} k(v)} \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} =: C_2 \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)}.$$

Integrating over  $B_\Omega(x_0, r)$  and using  $|B_\Omega(x_0, r)| \leq |B(x_0, r)| = \omega_N r^N$ , where  $\omega_N$  is the volume of the unit ball in  $\mathbb{R}^N$ , gives

$$\mu(B_\Omega(x_0, r)) = \int_{B_\Omega(x_0, r)} \phi(u) dx \leq C_2 |B(x_0, r)| \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} \leq C_3 r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)},$$

with  $C_3 := C_2 \omega_N$ .

**Lower bound.** We construct a smaller Euclidean ball  $B_c$  contained in  $B_\Omega(x_0, r)$  on which  $\phi$  admits a uniform positive lower bound in terms of  $\prod_{v \in R_+} (L_{v,0} + r)^{2k(v)}$ .

To this end, we first choose

$$\lambda := 1 + \frac{L_*}{m_*} > 1, \quad \eta := \frac{1}{4(\sqrt{N_0} \lambda + 1)}, \quad \theta := \lambda \eta.$$

These parameters depend only on  $L_*$  and  $m_*$ , and satisfy

$$\theta > \eta > 0, \quad \sqrt{N_0} \theta + \eta = \eta(\sqrt{N_0} \lambda + 1) = \frac{1}{4} < 1.$$

Let  $\mathbf{1}_{N_0} := (1, \dots, 1) \in \mathbb{R}^{N_0}$  and define the center

$$x_c := (u_c, z_c) := (u_0 + \theta r \mathbf{1}_{N_0}, z_0), \quad B_c := B(x_c, \eta r).$$

We *claim* that  $B_c \subset B_\Omega(x_0, r)$ . To see this inclusion, we first apply the triangle inequality to see that for any  $x \in B_c$ ,

$$\|x - x_0\| \leq \|x - x_c\| + \|x_c - x_0\| \leq \eta r + \theta r \sqrt{N_0} \leq r,$$

since  $\sqrt{N_0} \theta + \eta < 1$ . Hence  $B_c \subset B(x_0, r)$ . Moreover, for any  $x = (u, z) \in B_c$ , we have  $\|u - u_c\| \leq \|x - x_c\| \leq \eta r$ . This implies that for each  $1 \leq i \leq N_0$ ,

$$u_i \geq (u_c)_i - \eta r = (u_0)_i + (\theta - \eta)r \geq 0,$$

since  $u_0 \in \mathbb{R}_+^{N_0}$  and  $\theta \geq \eta$ . Thus  $u \in \mathbb{R}_+^{N_0}$  and  $x \in \Omega$ . Therefore,  $B_c \subset B_\Omega(x_0, r)$ . This finishes the proof of the claim.

Next we obtain a uniform lower bound for  $L_v(u)$  on  $B_c$ . Write any  $x = (u, z) \in B_c$  as

$$u = u_0 + \theta r \mathbf{1}_{N_0} + \Delta_u, \quad z = z_0 + \Delta_z, \quad \|\Delta_u\|^2 + \|\Delta_z\|^2 \leq (\eta r)^2,$$

so in particular  $\|\Delta_u\| \leq \eta r$ . Then for all  $x = (u, z) \in B_c$  and all  $v \in R_+$ ,

$$\begin{aligned} L_v(u) &= a_v \cdot u = a_v \cdot u_0 + \theta r a_v \cdot \mathbf{1}_{N_0} + a_v \cdot \Delta_u \\ &\geq L_{v,0} + \theta r \sum_{i=1}^{N_0} a_{v,i} - \|a_v\| \|\Delta_u\| \geq L_{v,0} + r K_0 \geq c_0 (L_{v,0} + r). \end{aligned}$$

with

$$K_0 := \theta m_* - \eta L_* = \eta(\lambda m_* - L_*) = \eta m_* > 0, \quad c_0 := \min\{1, K_0\} \in (0, 1].$$

Consequently, for all  $u \in B_c$ ,

$$\phi(u) = \prod_{v \in R_+} L_v(u)^{2k(v)} \geq c_0^{2 \sum_{v \in R_+} k(v)} \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} =: C_4 \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)}.$$

This, in combination with the fact that  $B_c \subset B_\Omega(x_0, r)$ , yields

$$\mu(B_\Omega(x_0, r)) \geq \mu(B_c) = \int_{B_c} \phi(u) dx \geq C_4 |B_c| \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} = C_5 r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)},$$

where  $C_5 := C_4 \omega_N \eta^N > 0$ .

The second statement follows immediately from the first one. Indeed, by the first statement,

$$\mu(B_\Omega(x_0, 2r)) \leq C_3 (2r)^N \prod_{v \in R_+} (L_{v,0} + 2r)^{2k(v)} \leq C_3 2^{N+\mathcal{K}} r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} \leq C_3 C_5^{-1} 2^{N+\mathcal{K}} \mu(B_\Omega(x_0, r)),$$

where we set  $\mathcal{K} := 2 \sum_{v \in R_+} k(v)$ . Therefore,  $(\Omega, \|\cdot\|, \mu)$  is doubling with the doubling constant  $C_{\text{dbl}, \Omega} := C_3 C_5^{-1} 2^{N+\mathcal{K}}$ . This finishes the proof of Lemma 2.12.  $\square$

Let  $\Phi : \mathbb{R}^N \longrightarrow \mathbb{R}^N$  be the linear isomorphism given in Corollary 2.6 such that  $\Phi(\Gamma) = \Omega$ . Since  $\Phi$  is linear and invertible, there exist positive constants  $c_\Phi$  and  $C_\Phi$  such that for any  $x, y \in \mathbb{R}^N$ ,

$$(8) \quad c_\Phi \|x - y\| \leq \|\Phi(x) - \Phi(y)\| \leq C_\Phi \|x - y\|.$$

Denote by  $B_\Gamma(x, r) := \{y \in \Gamma : \|y - x\| < r\}$ . It follows from (8) that for any  $x_0 \in \Gamma$  and  $r > 0$ ,

$$(9) \quad B_\Omega(\Phi(x_0), c_\Phi r) \subset \Phi(B_\Gamma(x_0, r)) \subset B_\Omega(\Phi(x_0), C_\Phi r).$$

This, together with Lemma 2.12, yields

$$(10) \quad \mu(\Phi(B_\Gamma(x_0, r))) \simeq \mu(B_\Omega(\Phi(x_0), C_\Phi r)).$$

**Lemma 2.13.** *For all  $x_0 \in \Gamma$  and  $r > 0$ , one can find an axis-parallel rectangle  $R'$  on  $\Omega$  such that*

$$(11) \quad B_\Omega(\Phi(x_0), C_\Phi r) \subset R' \subset B_\Omega(\Phi(x_0), \sqrt{N} C_\Phi r).$$

Furthermore, we have

$$(12) \quad \mu(R') \simeq \mu(B_\Omega(\Phi(x_0), C_\Phi r)).$$

*Proof.* We write  $(u_0, z_0) := \Phi(x_0) \in \Omega$ . and define the axis-parallel rectangle

$$(13) \quad R' := \left( \prod_{i=1}^{N_0} [\max\{0, (u_0)_i - C_\Phi r\}, (u_0)_i + C_\Phi r] \right) \times \left( \prod_{j=1}^{N-N_0} [(z_0)_j - C_\Phi r, (z_0)_j + C_\Phi r] \right).$$

On the one hand, it follows immediately that  $B_\Omega(\Phi(x_0), C_\Phi r) \subset R'$ . On the other hand, since  $R'$  is a rectangle with half-side lengths bounded by  $C_\Phi r$  in each coordinate, we have

$$R' \subset B_\Omega(\Phi(x_0), \sqrt{N} C_\Phi r).$$

The second statement follows from the first one together with Lemma 2.12. The proof of Lemma 2.13 is finished.  $\square$

The following theorem demonstrates that  $(\Gamma, \|\cdot\|, dw)$  is a doubling metric measure space.

**Theorem 2.14.** *There exists a constant  $C \geq 1$  such that for every  $x \in \Gamma$  and every  $r > 0$ ,*

$$(14) \quad C^{-1} r^N \prod_{v \in R_+} (|\langle x, v \rangle| + r)^{2k(v)} \leq w(B_\Gamma(x, r)) \leq C r^N \prod_{v \in R_+} (|\langle x, v \rangle| + r)^{2k(v)}.$$

*In particular, there exists a constant  $C_{\text{dbl}, \Gamma} \geq 1$  such that for every  $x \in \Gamma$  and every  $r > 0$ ,*

$$w(B_\Gamma(x, 2r)) \leq C_{\text{dbl}, \Gamma} w(B_\Gamma(x, r)).$$

*Moreover, for all  $x \in \Gamma$  and  $r > 0$ , we have*

$$(15) \quad w(B_\Gamma(x, r)) \sim w(B(x, r)).$$

*Proof.* The main strategy of the proof is to transfer the ball estimates from  $(\Omega, \|\cdot\|, \mu)$  back to  $(\Gamma, \|\cdot\|, dw)$ . Fix  $x \in \Gamma$  and  $r > 0$ . It follows from the equality (4) that

$$(16) \quad w(B_\Gamma(x, r)) = \mu(\Phi(B_\Gamma(x, r))).$$

Applying  $\mu$  to the inclusions (9) and using (16) yields

$$(17) \quad \mu(B_\Omega(\Phi(x), c_\Phi r)) \leq w(B_\Gamma(x, r)) \leq \mu(B_\Omega(\Phi(x), C_\Phi r)).$$

Write  $\Phi(x) = (u, z) \in \Omega$ . For each  $v \in R_+$  recall the linear form  $L_{v,0} := \sum_{i=1}^{N_0} a_{v,i} u_i$ . Combining the inequality (17) with Lemma 2.12 and the equality (6), we obtain that

$$(18) \quad w(B_\Gamma(x, r)) \simeq r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} = r^N \prod_{v \in R_+} (|\langle x, v \rangle| + r)^{2k(v)}.$$

This proves (14). The second statement follows immediately from (14). The third statement follows from the inequalities (14) and (1). This completes the proof of Theorem 2.14.  $\square$

### 3. SPECTRAL ASYMPTOTIC FORMULA

**Definition 3.1.** For  $g \in G$  and  $j = 1, \dots, N$ , set

$$d_{g,j} := g^{-1}e_j, \quad f_g(x) := f(gx).$$

For  $\alpha \in R_+$ , define  $S_\alpha : \mathbb{C}^G \rightarrow \mathbb{C}^G$  by

$$(S_\alpha z)_g := z_{g\sigma_\alpha}, \quad g \in G.$$

For  $f \in \dot{W}^{1,N}(\mathbb{R}^N)$ ,  $x \in \Gamma_+$ , and  $\theta \in \mathbb{S}^{N-1}$ , define

$$\begin{aligned} \mathfrak{c}_{f,j}(x, \theta) &:= \text{diag}_{g \in G} \left( \langle d_{g,j}, \nabla f_g(x) \rangle - \langle d_{g,j}, \theta \rangle \langle \theta, \nabla f_g(x) \rangle \right) \\ &\quad + \sum_{\alpha \in R_+} k(\alpha) \text{diag}_{g \in G} \left( \frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle} (f_g(x) - f_{g\sigma_\alpha}(x)) \right) S_\alpha. \end{aligned}$$

For smooth  $f$ , the quotients in the second term are understood by their smooth extensions across the corresponding reflection hyperplanes. For  $f \in \dot{W}^{1,N}(\mathbb{R}^N)$ ,  $\mathfrak{c}_{f,j}$  is understood almost everywhere on  $\Gamma \times \mathbb{S}^{N-1}$ .

**Lemma 3.2.** Let  $f \in W_{\text{loc}}^{1,1}(\mathbb{R}^N)$ . For every  $u \in C_c^\infty(\mathbb{R}^N)$ ,

$$(1) \quad [T_{e_j}, M_f]u(x) = \partial_j f(x)u(x) + \sum_{\alpha \in R_+} k(\alpha) \langle \alpha, e_j \rangle \frac{f(x) - f(\sigma_\alpha x)}{\langle \alpha, x \rangle} u(\sigma_\alpha x)$$

for almost every  $x \in \mathbb{R}^N$ . If  $N > 1$  and  $f \in \dot{W}^{1,N}(\mathbb{R}^N)$ , then, for every  $\alpha \in R_+$ ,

$$(2) \quad \left\| \frac{f - f \circ \sigma_\alpha}{\langle \alpha, \cdot \rangle} \right\|_{L^N(\mathbb{R}^N, dx)} \lesssim_{N,G} \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}.$$

*Proof.* Pairing  $\alpha$  and  $-\alpha$  in the definition of  $T_{e_j}$ , and using  $k(-\alpha) = k(\alpha)$  and  $\sigma_{-\alpha} = \sigma_\alpha$ , we have

$$T_{e_j}u(x) = \partial_j u(x) + \sum_{\alpha \in R_+} k(\alpha) \langle \alpha, e_j \rangle \frac{u(x) - u(\sigma_\alpha x)}{\langle \alpha, x \rangle}.$$

Therefore,

$$[T_{e_j}, M_f]u(x) = \partial_j f(x)u(x) + \sum_{\alpha \in R_+} k(\alpha) \langle \alpha, e_j \rangle \frac{f(x) - f(\sigma_\alpha x)}{\langle \alpha, x \rangle} u(\sigma_\alpha x),$$

which proves (1).

Since  $R$  is normalized,

$$x - \sigma_\alpha x = \langle \alpha, x \rangle \alpha, \quad \|x - \sigma_\alpha x\| = \sqrt{2} |\langle \alpha, x \rangle|.$$

Hence, for almost every  $x \notin \alpha^\perp$ ,

$$\frac{|f(x) - f(\sigma_\alpha x)|}{|\langle \alpha, x \rangle|} = \sqrt{2} \mathcal{G}_{I, \sigma_\alpha} f(x).$$

By Lemma ?? (??),

$$\left\| \frac{f - f \circ \sigma_\alpha}{\langle \alpha, \cdot \rangle} \right\|_{L^N(\mathbb{R}^N, dx)} \lesssim_{N, G} \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}.$$

This completes the proof of Lemma 3.2.  $\square$

**Lemma 3.3.** *Let  $\Gamma_+$  be an open Weyl chamber, let  $\Gamma$  be its closure, and let  $R_+$  be the corresponding positive subsystem. Set*

$$\rho(x) := \varpi(x)^{1/2},$$

and define

$$\begin{aligned} U : L^2(\mathbb{R}^N, dw) &\longrightarrow L^2(\Gamma, dw; \mathbb{C}^G), & (Uu)_g(x) &:= u(gx), \\ V : L^2(\Gamma, dw; \mathbb{C}^G) &\longrightarrow L^2(\Gamma, dx; \mathbb{C}^G), & (Vh)(x) &:= \rho(x)h(x). \end{aligned}$$

Then  $U$  and  $V$  are unitary. If

$$\tilde{T}_j := VUT_{e_j}U^{-1}V^{-1}, \quad \tilde{P} := VU(-\Delta_{\text{Dunkl}})U^{-1}V^{-1},$$

then, on  $C_c^\infty(\Gamma_+; \mathbb{C}^G)$ ,

$$(3) \quad (\tilde{T}_j h)_g = \partial_{d_{g,j}} h_g - \sum_{\alpha \in R_+} k(\alpha) \frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle} h_{g\sigma_\alpha},$$

$$(4) \quad \tilde{P} = -\Delta_{\mathbb{C}^G} + |\nabla \log \rho|^2 I_{\mathbb{C}^G} - 2 \sum_{\alpha \in R_+} k(\alpha) \frac{S_\alpha}{\langle \alpha, x \rangle^2}.$$

Assume in addition that  $N > 1$ , and let

$$\tilde{Q} := VU(-\Delta_{\text{Dunkl}})^{-1/2}U^{-1}V^{-1},$$

where  $(-\Delta_{\text{Dunkl}})^{-1/2}$  is defined by the spectral calculus. Then, for every  $\chi \in C_c^\infty(\Gamma_+)$ , the operator  $M_\chi \tilde{Q} M_\chi$ , initially defined on its natural domain, admits a bounded extension to  $L^2(\Gamma, dx; \mathbb{C}^G)$ , and this extension is a classical matrix-valued pseudodifferential operator of order  $-1$ . Its principal symbol is

$$\sigma_{-1}(M_\chi \tilde{Q} M_\chi)(x, \xi) = \chi(x)^2 |\xi|^{-1} I_{\mathbb{C}^G}.$$

Consequently, if  $x \in \Gamma_+$ ,  $\xi \neq 0$ , and  $\chi \in C_c^\infty(\Gamma_+)$  is identically 1 in a neighborhood of  $x$ , then

$$(5) \quad \sigma_{-1}(M_\chi \tilde{Q} M_\chi)(x, \xi) = |\xi|^{-1} I_{\mathbb{C}^G}.$$

*Proof.* The Weyl chambers  $g\Gamma_+$ ,  $g \in G$ , are pairwise disjoint and cover  $\mathbb{R}^N$  up to the reflection hyperplanes; see [6, Theorem 2.12]. Since  $dw$  is  $G$ -invariant,

$$\begin{aligned} \|Uu\|_{L^2(\Gamma, dw; \mathbb{C}^G)}^2 &= \sum_{g \in G} \int_{\Gamma} |u(gx)|^2 dw(x) \\ &= \int_{\mathbb{R}^N} |u(x)|^2 dw(x). \end{aligned}$$

Conversely, for  $h = (h_g)_{g \in G} \in L^2(\Gamma, dw; \mathbb{C}^G)$ , define  $u(gx) := h_g(x)$  for  $x \in \Gamma_+$ . This determines  $u$  almost everywhere on  $\mathbb{R}^N$  and gives  $Uu = h$ . Hence  $U$  is unitary. Since  $dw = \rho^2 dx$  and  $\rho > 0$  on  $\Gamma_+$ ,  $V$  is unitary as well.

Pairing the roots  $\alpha$  and  $-\alpha$  in the definition of  $T_{e_j}$ , changing variables in the root system by  $g$ , and using the  $G$ -invariance of  $k$ , we obtain, for  $h \in C_c^\infty(\Gamma_+; \mathbb{C}^G)$ ,

$$(UT_{e_j}U^{-1}h)_g = \partial_{d_{g,j}} h_g + \sum_{\alpha \in R_+} k(\alpha) \frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle} (h_g - h_{g\sigma_\alpha}).$$

On  $\Gamma_+$ ,

$$\nabla \log \rho(x) = \sum_{\alpha \in R_+} k(\alpha) \frac{\alpha}{\langle \alpha, x \rangle},$$

and therefore

$$\rho \partial_{d_{g,j}} \rho^{-1} = \partial_{d_{g,j}} - \sum_{\alpha \in R_+} k(\alpha) \frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle}.$$

Conjugation by  $V$  cancels the diagonal terms and gives (3).

Similarly, pairing  $\alpha$  and  $-\alpha$  in the formula for the Dunkl Laplacian and using  $\|\alpha\|^2 = 2$ , we have

$$\begin{aligned} (U(-\Delta_{\text{Dunkl}})U^{-1}h)_g &= -\Delta h_g - 2 \sum_{\alpha \in R_+} k(\alpha) \frac{\partial_\alpha h_g}{\langle \alpha, x \rangle} \\ &\quad + 2 \sum_{\alpha \in R_+} k(\alpha) \frac{h_g - h_{g\sigma_\alpha}}{\langle \alpha, x \rangle^2}. \end{aligned}$$

Writing  $\phi = \log \rho$ ,

$$\rho(-\Delta - 2\nabla \phi \cdot \nabla) \rho^{-1} = -\Delta + |\nabla \phi|^2 + \Delta \phi,$$

while

$$\Delta \phi = - \sum_{\alpha \in R_+} k(\alpha) \frac{\|\alpha\|^2}{\langle \alpha, x \rangle^2} = -2 \sum_{\alpha \in R_+} \frac{k(\alpha)}{\langle \alpha, x \rangle^2}.$$

The latter term cancels the diagonal part of

$$2 \sum_{\alpha \in R_+} k(\alpha) \frac{I_{\mathbb{C}^G} - S_\alpha}{\langle \alpha, x \rangle^2},$$

and hence

$$\tilde{P} = -\Delta I_{\mathbb{C}^G} + |\nabla \log \rho|^2 I_{\mathbb{C}^G} - 2 \sum_{\alpha \in R_+} k(\alpha) \frac{S_\alpha}{\langle \alpha, x \rangle^2},$$

which proves (4).

Assume now that  $N > 1$ , fix  $\chi \in C_c^\infty(\Gamma_+)$ , and set

$$K := \text{supp } \chi.$$

On  $\Gamma_+$ , write

$$\tilde{P} = -\Delta_{\mathbb{C}^G} + W(x), \quad W(x) = |\nabla \log \rho(x)|^2 I_{\mathbb{C}^G} - 2 \sum_{\alpha \in R_+} k(\alpha) \frac{S_\alpha}{\langle \alpha, x \rangle^2}.$$

Thus  $W$  is a smooth Hermitian matrix-valued function on  $\Gamma_+$ .

Choose open sets

$$K \Subset \Omega_0 \Subset \Omega_1 \Subset \Gamma_+$$

and  $\eta \in C_c^\infty(\Gamma_+)$  such that

$$0 \leq \eta \leq 1, \quad \eta = 1 \quad \text{on a neighborhood of } \overline{\Omega_1}.$$

Choose a flat torus  $M$  containing  $\text{supp } \eta$  in one Euclidean coordinate chart. Since  $\eta$  is compactly supported in that chart, the differential operator  $M_\eta \tilde{P} M_\eta$  extends smoothly by zero to  $M$ . Define on  $L^2(M; \mathbb{C}^G)$

$$(6) \quad P_0 := M_\eta \tilde{P} M_\eta + M_{1-\eta}(-\Delta_M + 1)M_{1-\eta}.$$

It is a nonnegative self-adjoint elliptic differential operator of order 2. Indeed,

$$\langle P_0 u, u \rangle = \langle \tilde{P}(\eta u), \eta u \rangle + \|\nabla((1-\eta)u)\|_2^2 + \|(1-\eta)u\|_2^2 \geq 0,$$

and its principal symbol is

$$p_{0,2}(x, \xi) = (\eta(x)^2 + (1-\eta(x))^2)|\xi|^2 I_{\mathbb{C}^G},$$

which is elliptic since

$$\eta^2 + (1-\eta)^2 \geq \frac{1}{2}.$$

Moreover,

$$(7) \quad P_0 = \tilde{P} \quad \text{on } \Omega_1.$$

Let

$$H_t := e^{-t\tilde{P}}, \quad H_t^0 := e^{-tP_0}.$$

We first consider

$$B_0 := \frac{1}{\sqrt{\pi}} \int_0^1 t^{-1/2} H_t^0 dt.$$

Let  $\Pi_0$  be the orthogonal projection onto  $\ker P_0$ . Since  $P_0$  is elliptic on the closed manifold  $M$ ,  $\Pi_0$  is smoothing. Applying the complex powers calculus [?, Theorem 2 and p. 290] to  $P_0 + \Pi_0$ , we obtain

$$(P_0 + \Pi_0)^{-1/2} \in \Psi_{\text{cl}}^{-1}(M; \mathbb{C}^G)$$

with principal symbol  $p_{0,2}^{-1/2}$ .

Set

$$F(\lambda) := \frac{1}{\sqrt{\pi}} \int_0^1 t^{-1/2} e^{-t\lambda} dt, \quad \lambda \geq 0.$$

Then  $B_0 = F(P_0)$ , and for  $\lambda > 0$ ,

$$F(\lambda) - \lambda^{-1/2} = -\frac{1}{\sqrt{\pi}} \int_1^\infty t^{-1/2} e^{-t\lambda} dt.$$

Hence

$$F(\lambda) - \lambda^{-1/2} = O(\lambda^{-m}) \quad (\lambda \rightarrow \infty)$$

for every  $m > 0$ . On  $\ker P_0$  the difference between  $B_0$  and  $(P_0 + \Pi_0)^{-1/2}$  is finite rank. It follows from elliptic spectral regularity that

$$B_0 - (P_0 + \Pi_0)^{-1/2} \in \Psi^{-\infty}(M; \mathbb{C}^G).$$

Consequently,

$$(8) \quad B_0 \in \Psi_{\text{cl}}^{-1}(M; \mathbb{C}^G), \quad \sigma_{-1}(B_0)(x, \xi) = p_{0,2}(x, \xi)^{-1/2}.$$

In particular, by (7),

$$(9) \quad \sigma_{-1}(B_0)(x, \xi) = |\xi|^{-1} I_{\mathbb{C}^G}, \quad x \in \Omega_1.$$

We next compare  $H_t$  and  $H_t^0$  in the interior. Choose  $\zeta \in C_c^\infty(\Omega_1)$  such that

$$\zeta = 1 \quad \text{on a neighborhood of } K.$$

Since  $P_0 = \tilde{P}$  on  $\Omega_1$ , Duhamel's formula gives

$$(10) \quad M_\chi(H_t - H_t^0)M_\chi = - \int_0^t M_\chi H_{t-s}[\tilde{P}, M_\zeta]H_s^0 M_\chi ds.$$

Indeed, for data supported in  $K$ ,

$$v(t) := M_\zeta H_t^0 M_\chi u$$

satisfies

$$(\partial_t + \tilde{P})v = [\tilde{P}, M_\zeta]H_t^0 M_\chi u, \quad v(0) = M_\chi u,$$

which yields (10).

Let  $h_t(x, y)$  be the Dunkl heat kernel. The integral kernel of  $H_t$ , with respect to Lebesgue measure on  $\Gamma$ , is the matrix

$$(11) \quad \tilde{h}_t(x, y)_{g, \ell} = \rho(x)\rho(y)h_t(gx, \ell y), \quad g, \ell \in G.$$

For  $x, y \in \Gamma$ ,

$$d(gx, \ell y) = d(x, y) = |x - y|.$$

Together with the mixed derivative estimate for the Dunkl heat kernel [?, Theorem 3.1], this implies that, whenever  $E, F \Subset \Gamma_+$  and  $\text{dist}(E, F) > 0$ , for every pair of multi-indices  $\alpha, \beta$  and every  $m > 0$ ,

$$(12) \quad \sup_{\substack{x \in E \\ y \in F}} \left\| \partial_x^\alpha \partial_y^\beta \tilde{h}_t(x, y) \right\| = O(t^m), \quad t \downarrow 0.$$

The heat kernel of the smooth elliptic operator  $P_0$  satisfies the same off-diagonal estimate on  $M$ .

Since

$$\text{supp}[\tilde{P}, M_\zeta] \cap K = \emptyset,$$

the two heat kernels occurring in (10) are evaluated between compact sets separated by a positive distance. The standard Gaussian bounds underlying (12) therefore give, for every multi-indices  $\alpha, \beta$  and every  $m > 0$ ,

$$(13) \quad \sup_{x, y} \left\| \partial_x^\alpha \partial_y^\beta K_{M_\chi(H_t - H_t^0)M_\chi}(x, y) \right\| = O(t^m), \quad t \downarrow 0.$$

Hence

$$(14) \quad S_0 := \frac{1}{\sqrt{\pi}} \int_0^1 t^{-1/2} M_\chi (H_t - H_t^0) M_\chi dt \in \Psi^{-\infty}.$$

It remains to treat the large-time part. Let  $K_1 \Subset \Gamma_+$  be a compact set containing  $K$ . By (11), the heat-kernel derivative estimate cited above, and (2), for  $x, y \in K_1$  and  $t \geq 1$ ,

$$V(gx, \ell y, \sqrt{t})^{-1} \lesssim_{K_1} t^{-N/2}.$$

Since  $\rho$  and all its derivatives are bounded on  $K_1$ , it follows that, for every pair of multi-indices  $\alpha, \beta$ ,

$$(15) \quad \sup_{x, y \in K_1} \left\| \partial_x^\alpha \partial_y^\beta \tilde{h}_t(x, y) \right\| \lesssim_{\alpha, \beta, K_1} t^{-N/2}, \quad t \geq 1.$$

Since  $N > 1$ ,

$$\int_1^\infty t^{-1/2} t^{-N/2} dt < \infty.$$

Thus all derivatives of

$$K_{S_\infty}(x, y) := \frac{\chi(x)\chi(y)}{\sqrt{\pi}} \int_1^\infty t^{-1/2} \tilde{h}_t(x, y) dt$$

converge locally uniformly, and therefore

$$(16) \quad S_\infty := \frac{1}{\sqrt{\pi}} \int_1^\infty t^{-1/2} M_\chi H_t M_\chi dt \in \Psi^{-\infty}.$$

For completeness, the preceding integrals give the canonical bounded localization of  $\tilde{P}^{-1/2}$ . Indeed, for  $\varepsilon > 0$ ,

$$(\tilde{P} + \varepsilon)^{-1/2} = \frac{1}{\sqrt{\pi}} \int_0^\infty t^{-1/2} e^{-\varepsilon t} H_t dt$$

by the spectral theorem. The estimates (13) and (15) imply that

$$M_\chi (\tilde{P} + \varepsilon)^{-1/2} M_\chi$$

converges in operator norm, as  $\varepsilon \downarrow 0$ , to

$$\frac{1}{\sqrt{\pi}} \int_0^\infty t^{-1/2} M_\chi H_t M_\chi dt.$$

On the natural domain of  $\tilde{P}^{-1/2}$ , this limit agrees with  $M_\chi \tilde{P}^{-1/2} M_\chi$ ; hence it is its bounded extension.

Combining (14), (16), and the definition of  $B_0$ , we obtain

$$M_\chi \tilde{Q} M_\chi = M_\chi B_0 M_\chi + S_0 + S_\infty.$$

By (8),  $M_\chi B_0 M_\chi$  is a classical matrix-valued pseudodifferential operator of order  $-1$ , while  $S_0 + S_\infty$  is smoothing. Therefore

$$M_\chi \tilde{Q} M_\chi \in \Psi_{\text{cl}}^{-1},$$

and, by (9),

$$\sigma_{-1}(M_\chi \tilde{Q} M_\chi)(x, \xi) = \chi(x)^2 |\xi|^{-1} I_{\mathbb{C}^G}.$$

If  $\chi = 1$  in a neighborhood of  $x$ , this reduces to

$$\sigma_{-1}(M_\chi \tilde{Q} M_\chi)(x, \xi) = |\xi|^{-1} I_{\mathbb{C}^G},$$

which proves (5).  $\square$

**Lemma 3.4.** *Let  $j \in \{1, \dots, N\}$  and  $f \in C_c^\infty(\mathbb{R}^N)$ , and set*

$$A_{f,j} := VU[R_{\text{Dunkl},j}, M_f]U^{-1}V^{-1}.$$

*Then, for every  $\chi \in C_c^\infty(\Gamma_+)$ ,*

$$M_\chi A_{f,j} M_\chi$$

*is a compactly supported classical matrix valued pseudodifferential operator of order  $-1$ , with principal symbol*

$$(17) \quad \sigma_{-1}(M_\chi A_{f,j} M_\chi)(x, \xi) = \frac{\chi(x)^2}{\|\xi\|} \mathbf{c}_{f,j}\left(x, \frac{\xi}{\|\xi\|}\right), \quad x \in \Gamma_+, \quad \xi \neq 0.$$

*Proof.* Set

$$F(x) := \text{diag}_{g \in G}(f_g(x)).$$

Since  $VUM_fU^{-1}V^{-1} = M_F$  and  $R_{\text{Dunkl},j} = T_{e_j}(-\Delta_{\text{Dunkl}})^{-1/2}$ ,

$$(18) \quad A_{f,j} = [\tilde{T}_j, M_F]\tilde{Q} + \tilde{T}_j[\tilde{Q}, M_F].$$

By (3),

$$(19) \quad \begin{aligned} [\tilde{T}_j, M_F] &= \text{diag}_{g \in G}(\langle d_{g,j}, \nabla f_g \rangle) \\ &+ \sum_{\alpha \in R_+} k(\alpha) \text{diag}_{g \in G} \left( \frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle} (f_g - f_{g\sigma_\alpha}) \right) S_\alpha. \end{aligned}$$

Fix  $\chi \in C_c^\infty(\Gamma_+)$ . Choose  $\eta, \psi \in C_c^\infty(\Gamma_+)$  such that  $\eta = 1$  on a neighborhood of  $\text{supp } \chi$  and  $\psi = 1$  on a neighborhood of  $\text{supp } \eta$ , and set

$$Q_\psi := M_\psi \tilde{Q} M_\psi.$$

By Lemma 3.3,  $Q_\psi$  is a classical matrix valued pseudodifferential operator of order  $-1$ , and on a neighborhood of  $\text{supp } \eta$ ,

$$(20) \quad q_{-1}(x, \xi) = \|\xi\|^{-1} I_{\mathbb{C}^G}.$$

We use the Kohn–Nirenberg convention

$$\text{Op}(a)u(x) = (2\pi)^{-N} \int_{\mathbb{R}^{2N}} e^{i(x-y) \cdot \xi} a(x, \xi) u(y) dy d\xi.$$

If  $a$  and  $b$  are classical symbols, the composition formula [?, Theorem 18.1.8, p. 71] gives

$$(21) \quad a \# b \sim \sum_{\beta \in \mathbb{N}_0^N} \frac{1}{\beta!} \partial_\xi^\beta a D_x^\beta b, \quad D_x := \frac{1}{i} \partial_x.$$

Write, on a neighborhood of  $\text{supp } \eta$ ,

$$\sigma(Q_\psi)(x, \xi) \sim q_{-1}(x, \xi) + q_{-2}(x, \xi) + \dots$$

By (4),

$$\sigma(\tilde{P})(x, \xi) \sim p_2(x, \xi) + p_1(x, \xi) + p_0(x, \xi),$$

where

$$p_2(x, \xi) = \|\xi\|^2 I_{\mathbb{C}^G}, \quad p_1(x, \xi) = 0.$$

Since  $\tilde{Q} = \tilde{P}^{-1/2}$ ,  $\tilde{Q}\tilde{P}\tilde{Q} = I$ . Applying (21) on the region where  $\psi = 1$  and comparing the homogeneous terms of degree  $-1$ , the terms containing  $x$ -derivatives of  $p_2$  and  $q_{-1}$  vanish, and hence

$$q_{-2}p_2q_{-1} + q_{-1}p_2q_{-2} = 0.$$

Together with (20), this gives

$$(22) \quad q_{-2}(x, \xi) = 0$$

on a neighborhood of  $\text{supp } \eta$ .

Applying (21) to the commutator  $[Q_\psi, M_F]$ , and using (20)–(22), we obtain, on the same neighborhood,

$$(23) \quad \begin{aligned} \sigma_{-2}([Q_\psi, M_F])(x, \xi) &= \frac{1}{i} \sum_{\ell=1}^N \partial_{\xi_\ell} (\|\xi\|^{-1}) \partial_{x_\ell} F(x) \\ &= i \|\xi\|^{-3} \text{diag}_{g \in G} (\langle \xi, \nabla f_g(x) \rangle). \end{aligned}$$

Since  $\psi = 1$  on a neighborhood of  $\text{supp } \eta$  and  $\eta = 1$  on a neighborhood of  $\text{supp } \chi$ ,

$$\begin{aligned} M_\chi[\tilde{T}_j, M_F]\tilde{Q}M_\chi &= M_\chi[\tilde{T}_j, M_F]Q_\psi M_\chi, \\ M_\chi\tilde{T}_j[\tilde{Q}, M_F]M_\chi &= M_\chi\tilde{T}_jM_\eta[Q_\psi, M_F]M_\chi. \end{aligned}$$

Both operators on the right are classical matrix valued pseudodifferential operators of order  $-1$ .

By (19), (20), and the order-zero part of the composition formula,

$$(24) \quad \begin{aligned} \sigma_{-1}(M_\chi[\tilde{T}_j, M_F]\tilde{Q}M_\chi)(x, \xi) &= \frac{\chi(x)^2}{\|\xi\|} \text{diag}_{g \in G} (\langle d_{g,j}, \nabla f_g(x) \rangle) \\ &+ \frac{\chi(x)^2}{\|\xi\|} \sum_{\alpha \in R_+} k(\alpha) \text{diag}_{g \in G} \left( \frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle} (f_g(x) - f_{g\sigma_\alpha}(x)) \right) S_\alpha. \end{aligned}$$

Under the above Kohn–Nirenberg convention, (3) gives

$$\sigma_1(\tilde{T}_j)(x, \xi) = i \text{diag}_{g \in G} (\langle d_{g,j}, \xi \rangle).$$

Therefore, by (23),

$$(25) \quad \begin{aligned} \sigma_{-1}(M_\chi\tilde{T}_j[\tilde{Q}, M_F]M_\chi)(x, \xi) &= -\chi(x)^2 \|\xi\|^{-3} \text{diag}_{g \in G} (\langle d_{g,j}, \xi \rangle \langle \xi, \nabla f_g(x) \rangle). \end{aligned}$$

Combining (24) and (25), and setting

$$\theta := \frac{\xi}{\|\xi\|},$$

we obtain

$$\begin{aligned} \sigma_{-1}(M_\chi A_{f,j} M_\chi)(x, \xi) &= \frac{\chi(x)^2}{\|\xi\|} \text{diag}_{g \in G} \left( \langle d_{g,j}, \nabla f_g(x) \rangle - \langle d_{g,j}, \theta \rangle \langle \theta, \nabla f_g(x) \rangle \right) \\ &+ \frac{\chi(x)^2}{\|\xi\|} \sum_{\alpha \in R_+} k(\alpha) \text{diag}_{g \in G} \left( \frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle} (f_g(x) - f_{g\sigma_\alpha}(x)) \right) S_\alpha \\ &= \frac{\chi(x)^2}{\|\xi\|} \mathbf{c}_{f,j}(x, \theta). \end{aligned}$$

This proves (17).  $\square$

**Lemma 3.5.** *Assume that  $N > 1$ ,  $j \in \{1, \dots, N\}$ , and  $f \in \dot{W}^{1,N}(\mathbb{R}^N)$ . Then*

$$\left( \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \operatorname{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N} \simeq_{N,R,k} \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}.$$

*Proof.* By the definition of  $\mathbf{c}_{f,j}$ , the unitary invariance of the Schatten norm, and  $\|d_{g,j}\| = 1$ ,

$$\begin{aligned} \|\mathbf{c}_{f,j}(x, \theta)\|_{S^N} &\lesssim_{R,k} \left( \sum_{g \in G} |\nabla f_g(x)|^N \right)^{1/N} \\ &\quad + \sum_{\alpha \in R_+} \left( \sum_{g \in G} \left| \frac{f_g(x) - f_{g\sigma_\alpha}(x)}{\langle \alpha, x \rangle} \right|^N \right)^{1/N}. \end{aligned}$$

The chamber decomposition used in Lemma 3.3 and the orthogonality of  $g \in G$  imply

$$\sum_{g \in G} \int_{\Gamma} |\nabla f_g(x)|^N dx = \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}^N.$$

Moreover, since  $\|\alpha\|^2 = 2$ ,

$$|x - \sigma_\alpha x| = \sqrt{2} |\langle \alpha, x \rangle|,$$

and hence

$$\left| \frac{f_g(x) - f_{g\sigma_\alpha}(x)}{\langle \alpha, x \rangle} \right| = \sqrt{2} \mathcal{G}_{g, g\sigma_\alpha} f(x)$$

for almost every  $x \in \mathbb{R}^N$ . By Lemma ??,

$$\int_{\Gamma} \left| \frac{f_g(x) - f_{g\sigma_\alpha}(x)}{\langle \alpha, x \rangle} \right|^N dx \lesssim_{N,G} \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}^N$$

uniformly in  $g \in G$  and  $\alpha \in R_+$ . Since  $G$  and  $R_+$  are finite,

$$(26) \quad \left( \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \operatorname{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N} \lesssim_{N,R,k} \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}.$$

For the converse estimate, let  $P_g$  denote the orthogonal projection of  $\mathbb{C}^G$  onto the  $g$ -th coordinate and define

$$\mathcal{E}(A) := \sum_{g \in G} P_g A P_g, \quad A \in \mathcal{L}(\mathbb{C}^G).$$

Thus,  $\mathcal{E}(A)$  is the diagonal part of  $A$ . The pinching inequality for unitarily invariant norms [?, p. 97, (IV.51)–(IV.52)] gives

$$(27) \quad \|\mathcal{E}(A)\|_{S^N} \leq \|A\|_{S^N}.$$

For every  $\alpha \in R_+$ ,  $S_\alpha$  has zero diagonal, since  $g\sigma_\alpha \neq g$  for all  $g \in G$ . Hence, by the definition of  $\mathbf{c}_{f,j}$ ,

$$\mathcal{E}(\mathbf{c}_{f,j}(x, \theta)) = \operatorname{diag}_{g \in G} \left( \langle d_{g,j}, \nabla f_g(x) \rangle - \langle d_{g,j}, \theta \rangle \langle \theta, \nabla f_g(x) \rangle \right).$$

Therefore, by (27),

$$(28) \quad \|\mathbf{c}_{f,j}(x, \theta)\|_{S^N}^N \geq \sum_{g \in G} |\langle d_{g,j}, \nabla f_g(x) \rangle - \langle d_{g,j}, \theta \rangle \langle \theta, \nabla f_g(x) \rangle|^N.$$

For each  $g \in G$ , set

$$y := gx, \quad \eta := g\theta.$$

Since

$$\nabla f_g(x) = g^{-1} \nabla f(y), \quad d_{g,j} = g^{-1} e_j,$$

and  $g$  is orthogonal,

$$\begin{aligned} \langle d_{g,j}, \nabla f_g(x) \rangle &= \partial_j f(y), \\ \langle d_{g,j}, \theta \rangle &= \eta_j, \\ \langle \theta, \nabla f_g(x) \rangle &= \langle \eta, \nabla f(y) \rangle. \end{aligned}$$

Using the chamber decomposition and the orthogonal invariance of surface measure, (28) therefore gives

$$(29) \quad \begin{aligned} & \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \|\mathbf{c}_{f,j}(x, \theta)\|_{S^N}^N d\theta dx \\ & \geq \int_{\mathbb{R}^N} \int_{\mathbb{S}^{N-1}} |\partial_j f(y) - \eta_j \langle \eta, \nabla f(y) \rangle|^N d\eta dy. \end{aligned}$$

For  $v \in \mathbb{C}^N$ , set

$$\Phi_j(v) := \left( \int_{\mathbb{S}^{N-1}} |v_j - \eta_j \langle \eta, v \rangle|^N d\eta \right)^{1/N}.$$

Then

$$(30) \quad \Phi_j(v) \simeq_N \|v\|.$$

Indeed,  $\Phi_j$  is a seminorm on  $\mathbb{C}^N$ . If  $\Phi_j(v) = 0$ , continuity gives

$$v_j - \eta_j \langle \eta, v \rangle = 0, \quad \eta \in \mathbb{S}^{N-1}.$$

Since  $N > 1$ , there exists  $\eta \in \mathbb{S}^{N-1}$  with  $\eta_j = 0$ , and therefore  $v_j = 0$ . It follows that

$$\langle \eta, v \rangle = 0$$

whenever  $\eta_j \neq 0$ . Since such  $\eta$  form a dense subset of  $\mathbb{S}^{N-1}$ , continuity yields  $\langle \eta, v \rangle = 0$  for every  $\eta \in \mathbb{S}^{N-1}$ , and hence  $v = 0$ . Thus  $\Phi_j$  is a norm, and (30) follows from the equivalence of norms on the finite dimensional space  $\mathbb{C}^N$ .

Applying (30) with  $v = \nabla f(y)$  in (29), we obtain

$$\|\nabla f\|_{L^N(\mathbb{R}^N, dx)} \lesssim_N \left( \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N}.$$

Combining this estimate with (26) proves the assertion.  $\square$

**Lemma 3.6.** *Assume that  $N > 1$ ,  $j \in \{1, \dots, N\}$ , and  $f \in C_c^\infty(\mathbb{R}^N)$ . Define*

$$A_f := VU[R_{\text{Dunkl},j}, M_f]U^{-1}V^{-1}.$$

*Then there exist  $\chi_m \in C_c^\infty(\Gamma_+)$ ,  $0 \leq \chi_m \leq 1$ , such that  $\chi_m(x) \rightarrow 1$  for every  $x \in \Gamma_+$  and*

$$(31) \quad \lim_{m \rightarrow \infty} \|A_f - M_{\chi_m} A_f M_{\chi_m}\|_{S^{N,\infty}} = 0.$$

*Proof.* Fix  $s > 2$ , choose  $2 < s_0 < s$ , and set

$$\gamma := \frac{1}{s_0} - \frac{1}{s} > 0.$$

Let  $r_s > s$  be as in Lemma ?? and set  $\beta := N/r_s$ . By Lemma ?? (??),  $f \in OSC^{N,\infty}(\mathbb{R}^N)$ , so Lemma ?? applies with  $p = N$  and  $q = \infty$ .

Set

$$B_f := U[R_{\text{Dunkl},j}, M_f]U^{-1}.$$

Since  $V = M_\rho$  commutes with scalar multiplication operators on  $\Gamma$ ,

$$(32) \quad \|A_f - M_\chi A_f M_\chi\|_{S^{N,\infty}} = \|B_f - M_\chi B_f M_\chi\|_{S^{N,\infty}}$$

for every bounded scalar multiplier  $\chi$ .

For fixed  $\sigma, \tau \in G$ ,  $u, v \in \mathbb{Z}^N$ , and  $\varepsilon \in \{0, 1, 2, 3\}$ , define

$$T_{u,v,\varepsilon}^{\sigma,\tau} := \sum_{I \in \mathbb{D}_\Gamma} A_{\sigma,\tau,I,\varepsilon}(f) \left\langle \cdot, \overline{H_{u,v,I,\varepsilon}^{\sigma,\tau}(\tau \cdot)} \right\rangle_{L^2(\Gamma, dw)} E_{u,v,I,\varepsilon}^{\sigma,\tau}(\sigma \cdot).$$

By the operator series identification in the proof of Theorem ??, the  $(\sigma, \tau)$  matrix entry of  $B_f$  is

$$\sum_{\varepsilon=0}^3 \sum_{u,v \in \mathbb{Z}^N} c_{u,v} T_{u,v,\varepsilon}^{\sigma,\tau},$$

with convergence in  $S^{N,\infty}$ . Moreover, by Lemma ?? (??)–(??),

$$(33) \quad \text{supp } E_{u,v,I,\varepsilon}^{\sigma,\tau}(\sigma \cdot) \subset I, \quad \text{supp } H_{u,v,I,\varepsilon}^{\sigma,\tau}(\tau \cdot) \subset I^*,$$

(34)

$$\|E_{u,v,I,\varepsilon}^{\sigma,\tau}(\sigma \cdot)\|_{L^s(dw)} + \|H_{u,v,I,\varepsilon}^{\sigma,\tau}(\tau \cdot)\|_{L^s(dw)} \lesssim w(I)^{1/s-1/2}.$$

The definitions of  $A_{\sigma,\tau,I,\varepsilon}(f)$  in the proof of Lemma ?? imply

$$(35) \quad |A_{\sigma,\tau,I,\varepsilon}(f)| \lesssim_f \begin{cases} \ell(I), & \ell(I) \leq 1, \\ \ell(I)^{-\beta}, & \ell(I) \geq 1. \end{cases}$$

Indeed, for  $\varepsilon = 0, 1, 2$ , the corresponding coefficients are bounded by an  $L^{r_s}(dx)$  oscillation of a smooth compactly supported function on a set of diameter comparable with  $\ell(I)$ . Hence the first estimate follows from the Lipschitz bound, while for  $\ell(I) \geq 1$ ,

$$\text{osc}_{r_s}(g; E) \lesssim |E|^{-1/r_s} \|g\|_{L^{r_s}} \lesssim_f \ell(I)^{-N/r_s}.$$

For  $\varepsilon = 3$ , writing  $D := |\sigma z_I - \tau z_I|$ , the definitions of  $\theta_{\sigma,\tau,I}$  and  $a_{\sigma,\tau,I}(f)$  imply, for  $\ell(I) \leq 1$ ,

$$|A_{\sigma,\tau,I,3}(f)| \lesssim_f \frac{\ell(I)}{\ell(I) + D} (\ell(I) + D) \lesssim_f \ell(I),$$

whereas, for  $\ell(I) \geq 1$ ,

$$|A_{\sigma,\tau,I,3}(f)| \leq |I|^{-1} \|H_{\sigma,\tau}(f)\|_{L^1} \lesssim_f \ell(I)^{-N} \lesssim \ell(I)^{-\beta}.$$

Let  $\Delta = \{\alpha_1, \dots, \alpha_{N_0}\}$ . For  $1 \leq i \leq N_0$  and  $0 < \delta \leq 1$ , set

$$W_{i,\delta} := \{x \in \Gamma : 0 \leq \langle x, \alpha_i \rangle \leq 2\delta\}.$$

We have

$$(36) \quad \frac{w(W_{i,\delta} \cap I)}{w(I)} + \frac{w(W_{i,\delta} \cap I^*)}{w(I)} \lesssim \min\left(1, \frac{\delta}{\ell(I)}\right).$$

To prove this, write  $I = \Phi^{-1}(Q)$  and  $\ell := \ell(I)$ . The assertion is immediate when  $\delta > \ell/8$ , since  $w(I^*) \lesssim w(I)$ . Suppose that  $\delta \leq \ell/8$ . If  $I$  or  $I^*$  meets  $W_{i,\delta}$ , then the  $i$ -th coordinate interval of  $Q$  is  $[0, \ell)$ . By Lemma 2.9 (5), the density of  $\Phi_{\#}dw$  is

$$\phi(u) = |\det(\Phi)|^{-1} \prod_{v \in R_+} \left( \sum_{h=1}^{N_0} a_{v,h} u_h \right)^{2k(v)},$$

which is nondecreasing in every  $u_i$ . If  $g$  is nonnegative and nondecreasing on an interval of length comparable with  $\ell$ , then

$$\int_0^{2\delta} g(t) dt \lesssim \frac{\delta}{\ell} \int g(t) dt.$$

Applying this estimate fiberwise to  $\phi$  on  $Q$ , respectively on  $(2Q) \cap \Omega$ , and using  $w(I^*) \lesssim w(I)$ , we obtain (36).

Define

$$\omega_{I,\delta} := \min\left(1, \left(\frac{\delta}{\ell(I)}\right)^\gamma\right).$$

If  $0 \leq a \leq 1$  and  $\text{supp } a \subset W_{i,\delta}$ , then by Hölder's inequality, (34), and (36),

$$\|aE_{u,v,I,\varepsilon}^{\sigma,\tau}(\sigma \cdot)\|_{L^{s_0}(dw)} + \|aH_{u,v,I,\varepsilon}^{\sigma,\tau}(\tau \cdot)\|_{L^{s_0}(dw)} \lesssim \omega_{I,\delta} w(I)^{1/s_0-1/2}.$$

After normalization by  $\omega_{I,\delta}$ , the localized factors are NWO families with exponent  $s_0$ . Hence, by Lemma ??,

$$(37) \quad \|M_a T_{u,v,\varepsilon}^{\sigma,\tau}\|_{S^{N,\infty}} + \|T_{u,v,\varepsilon}^{\sigma,\tau} M_a\|_{S^{N,\infty}} \lesssim \left\| \left\{ A_{\sigma,\tau,I,\varepsilon}(f) \omega_{I,\delta} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{\ell^{N,\infty}}.$$

If  $\ell(I) \leq 1$  and  $A_{\sigma,\tau,I,\varepsilon}(f) \neq 0$ , then  $z_I$  belongs to a fixed bounded set depending only on  $f$ . Consequently, at every dyadic length  $0 < \ell \leq 1$ ,

$$(38) \quad \#\left\{ I \in \mathbb{D}_\Gamma : \ell(I) = \ell, A_{\sigma,\tau,I,\varepsilon}(f) \neq 0, (I \cup I^*) \cap W_{i,\delta} \neq \emptyset \right\} \lesssim_f \left(1 + \frac{\delta}{\ell}\right) \ell^{-(N-1)}.$$

Indeed, under  $\Phi$ , there are  $O(1 + \delta/\ell)$  admissible intervals in the  $u_i$  coordinate and  $O(\ell^{-1})$  admissible intervals in each of the remaining  $N - 1$  coordinates.

For  $\ell(I) \leq \delta$ , (35) and (38) imply, for every  $\lambda > 0$ ,

$$\begin{aligned} & \# \left\{ I : \ell(I) \leq \delta, |A_{\sigma,\tau,I,\varepsilon}(f)\omega_{I,\delta}| > \lambda \right\} \\ & \lesssim_f \delta \sum_{\substack{\ell \leq \delta \\ \lambda \lesssim_f \ell}} \ell^{-N} \lesssim_f \delta \lambda^{-N}. \end{aligned}$$

Therefore,

$$(39) \quad \left\| \left\{ A_{\sigma,\tau,I,\varepsilon}(f)\omega_{I,\delta} \right\}_{\ell(I) \leq \delta} \right\|_{\ell^{N,\infty}} \lesssim_f \delta^{1/N}.$$

For  $\delta < \ell(I) \leq 1$ ,

$$|A_{\sigma,\tau,I,\varepsilon}(f)\omega_{I,\delta}| \lesssim_f \delta^\gamma \ell(I)^{1-\gamma},$$

and (38) reduces to  $O_f(\ell^{-(N-1)})$ . Given  $\lambda > 0$ , set

$$L := \max \left( \delta, c_f \left( \frac{\lambda}{\delta^\gamma} \right)^{1/(1-\gamma)} \right).$$

Since  $N > 1$ ,

$$\# \left\{ I : \delta < \ell(I) \leq 1, |A_{\sigma,\tau,I,\varepsilon}(f)\omega_{I,\delta}| > \lambda \right\} \lesssim_f L^{-(N-1)}.$$

If  $L = \delta$ , then  $\lambda \lesssim_f \delta$ , and hence

$$\lambda L^{-(N-1)/N} \lesssim_f \delta^{1/N}.$$

Otherwise,  $\lambda \lesssim_f \delta^\gamma L^{1-\gamma}$ , so

$$\lambda L^{-(N-1)/N} \lesssim_f \delta^\gamma L^{1/N-\gamma} \lesssim_f \delta^{\min(\gamma, 1/N)}.$$

Thus,

$$(40) \quad \left\| \left\{ A_{\sigma,\tau,I,\varepsilon}(f)\omega_{I,\delta} \right\}_{\delta < \ell(I) \leq 1} \right\|_{\ell^{N,\infty}} \lesssim_f \delta^{\min(\gamma, 1/N)}.$$

For  $\ell(I) \geq 1$ , only  $O_f(1)$  cubes at each dyadic scale have nonzero coefficients.

By (35),

$$\sum_{\ell(I) \geq 1} |A_{\sigma,\tau,I,\varepsilon}(f)\omega_{I,\delta}|^N \lesssim_f \delta^{\gamma N} \sum_{\substack{\ell \geq 1 \\ \text{dyadic}}} \ell^{-N(\beta+\gamma)} \lesssim_f \delta^{\gamma N}.$$

Combining this with (39) and (40), we obtain

$$(41) \quad \left\| \left\{ A_{\sigma,\tau,I,\varepsilon}(f)\omega_{I,\delta} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{\ell^{N,\infty}} \lesssim_f \delta^\kappa, \quad \kappa := \min \left( \gamma, \frac{1}{N} \right).$$

By (37),

$$(42) \quad \|M_a T_{u,v,\varepsilon}^{\sigma,\tau}\|_{S^{N,\infty}} + \|T_{u,v,\varepsilon}^{\sigma,\tau} M_a\|_{S^{N,\infty}} \lesssim_f \delta^\kappa$$

whenever  $0 \leq a \leq 1$  and  $\text{supp } a \subset W_{i,\delta}$ .

We also need localization at infinity. Let  $0 \leq b_R \leq 1$  with  $\text{supp } b_R \subset \{x \in \Gamma : |x| \geq R\}$ . If  $A_{\sigma,\tau,I,\varepsilon}(f) \neq 0$ , then

$$(43) \quad |z_I| \lesssim_f 1 + \ell(I).$$

For  $\varepsilon = 0, 1$ , the ball  $B_\Gamma(z_I, C_1 \ell(I))$  must meet the support of  $f \circ \tau$ ; for  $\varepsilon = 2, 3$ ,  $I$  must meet the support of  $H_{\sigma,\tau}(f)$ . Since  $|x - z_I| \lesssim \ell(I)$  for  $x \in I^*$ , (43) shows that, for sufficiently large  $R$ ,

$$A_{\sigma,\tau,I,\varepsilon}(f) \neq 0, \quad (I \cup I^*) \cap \{x : |x| \geq R\} \neq \emptyset$$

can occur only when  $\ell(I) \gtrsim_f R$ . At each such dyadic scale there are  $O_f(1)$  cubes with nonzero coefficients. Hence, by (35) and Lemma ??,

$$(44) \quad \|M_{b_R} T_{u,v,\varepsilon}^{\sigma,\tau}\|_{S^{N,\infty}} + \|T_{u,v,\varepsilon}^{\sigma,\tau} M_{b_R}\|_{S^{N,\infty}} \lesssim_f R^{-\beta}.$$

Choose  $\eta \in C^\infty(\mathbb{R})$ ,  $0 \leq \eta \leq 1$ , such that

$$\eta(t) = 0 \quad \text{for } t \leq 1, \quad \eta(t) = 1 \quad \text{for } t \geq 2,$$

and choose  $\zeta \in C_c^\infty(\mathbb{R}^N)$ ,  $0 \leq \zeta \leq 1$ , such that

$$\zeta = 1 \quad \text{on } B(0, 1), \quad \text{supp } \zeta \subset B(0, 2).$$

For  $0 < \delta \leq 1$  and  $R \geq 1$ , define

$$\chi_{\delta,R}(x) := \zeta(x/R) \prod_{i=1}^{N_0} \eta\left(\frac{\langle x, \alpha_i \rangle}{\delta}\right), \quad x \in \Gamma_+.$$

Then  $\chi_{\delta,R} \in C_c^\infty(\Gamma_+)$  and  $0 \leq \chi_{\delta,R} \leq 1$ . Moreover,

$$\begin{aligned} 1 - \chi_{\delta,R} &= 1 - \zeta(\cdot/R) \\ &+ \sum_{i=1}^{N_0} \zeta(\cdot/R) \prod_{h < i} \eta\left(\frac{\langle \cdot, \alpha_h \rangle}{\delta}\right) \left(1 - \eta\left(\frac{\langle \cdot, \alpha_i \rangle}{\delta}\right)\right). \end{aligned}$$

The first term is supported in  $\{x : |x| \geq R\}$ , while the  $i$ -th term in the sum is supported in  $W_{i,\delta}$ . Since  $G$  is finite,  $S^{N,\infty}$  is normable for  $N > 1$ , and Lemma ?? implies

$$\sum_{u,v \in \mathbb{Z}^N} |c_{u,v}| < \infty,$$

we may sum (42) and (44) over the operator series for  $B_f$ . Therefore,

$$\|(I - M_{\chi_{\delta,R}})B_f\|_{S^{N,\infty}} + \|B_f(I - M_{\chi_{\delta,R}})\|_{S^{N,\infty}} \lesssim_f \delta^\kappa + R^{-\beta}.$$

Since

$$B_f - M_{\chi_{\delta,R}} B_f M_{\chi_{\delta,R}} = (I - M_{\chi_{\delta,R}})B_f + M_{\chi_{\delta,R}} B_f (I - M_{\chi_{\delta,R}}),$$

the ideal property of  $S^{N,\infty}$  and (32) imply

$$\|A_f - M_{\chi_{\delta,R}} A_f M_{\chi_{\delta,R}}\|_{S^{N,\infty}} \lesssim_f \delta^\kappa + R^{-\beta}.$$

Define

$$\chi_m := \chi_{2^{-m}, 2^m}.$$

Then  $\chi_m(x) \rightarrow 1$  for every  $x \in \Gamma_+$ , while

$$\|A_f - M_{\chi_m} A_f M_{\chi_m}\|_{S^{N,\infty}} \lesssim_f 2^{-m\kappa} + 2^{-m\beta} \rightarrow 0.$$

This completes the proof of Lemma 3.6.  $\square$

**Lemma 3.7.** *Assume that  $N > 1$ ,  $j \in \{1, \dots, N\}$ , and  $f \in C_c^\infty(\mathbb{R}^N)$ . Then*

$$(45) \quad \lim_{t \rightarrow \infty} t^{1/N} \mu(t, [R_{\text{Dunkl},j}, M_f]) = \left( \frac{1}{N(2\pi)^N} \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N}.$$

*Proof.* Define

$$A_f := VU[R_{\text{Dunkl},j}, M_f]U^{-1}V^{-1}.$$

By Lemma 3.6 (31), there exist  $\chi_m \in C_c^\infty(\Gamma_+)$ ,  $0 \leq \chi_m \leq 1$ , such that  $\chi_m(x) \rightarrow 1$  for every  $x \in \Gamma_+$  and

$$(46) \quad \|A_f - M_{\chi_m} A_f M_{\chi_m}\|_{S^{N,\infty}} \rightarrow 0.$$

Set

$$A_{f,m} := M_{\chi_m} A_f M_{\chi_m}, \quad K_m := \text{supp } \chi_m.$$

By Lemma 3.4 (17),  $A_{f,m}$  is a compactly supported classical matrix valued pseudodifferential operator of order  $-1$  on  $\Gamma_+$ , with

$$(47) \quad \sigma_{-1}(A_{f,m})(x, \xi) = \frac{\chi_m(x)^2}{\|\xi\|} \mathbf{c}_{f,j}\left(x, \frac{\xi}{\|\xi\|}\right), \quad x \in \Gamma_+, \quad \xi \neq 0.$$

Choose a bounded open set  $U_m \Subset U_m \subseteq \Gamma_+$ , and choose  $L_m > 0$  such that

$$L_m > \sup_{x,y \in U_m} \|x - y\|_\infty.$$

For

$$\mathbb{T}_m^N := \mathbb{R}^N / L_m \mathbb{Z}^N,$$

the quotient map restricts to a diffeomorphism

$$\kappa_m : U_m \rightarrow \Omega_m := \kappa_m(U_m),$$

which is a local isometry. Since the Schwartz kernel of  $A_{f,m}$  is supported in  $K_m \times K_m$ , with respect to

$$L^2(\Gamma; \mathbb{C}^G) = L^2(U_m; \mathbb{C}^G) \oplus L^2(\Gamma \setminus U_m; \mathbb{C}^G)$$

we have

$$A_{f,m} = B_{f,m} \oplus 0$$

for some  $B_{f,m}$  on  $L^2(U_m; \mathbb{C}^G)$ .

Let

$$W_m : L^2(U_m; \mathbb{C}^G) \rightarrow L^2(\Omega_m; \mathbb{C}^G), \quad W_m u := u \circ \kappa_m^{-1}.$$

Then  $W_m$  is unitary. By the invariance of classical pseudodifferential operators under smooth coordinate changes [?, Theorem 18.1.17],

$$T_{f,m} := W_m B_{f,m} W_m^{-1} \in \Psi_{\text{cl}}^{-1}(\Omega_m; \mathbb{C}^G).$$

Its Schwartz kernel is supported in  $\kappa_m(K_m) \times \kappa_m(K_m) \subseteq \Omega_m \times \Omega_m$ . Choose  $\vartheta_m \in C_c^\infty(\Omega_m)$  equal to 1 on a neighborhood of  $\kappa_m(K_m)$ . Then

$$T_{f,m} = M_{\vartheta_m} T_{f,m} M_{\vartheta_m}.$$

Hence the zero extension of the Schwartz kernel of  $T_{f,m}$  from  $\Omega_m \times \Omega_m$  to  $\mathbb{T}_m^N \times \mathbb{T}_m^N$  is a classical pseudodifferential kernel of order  $-1$ : locally in  $\Omega_m \times \Omega_m$  this follows

from  $T_{f,m} \in \Psi_{\text{cl}}^{-1}$ , while near the boundary of  $\Omega_m \times \Omega_m$  the kernel vanishes. Denote the corresponding operator on  $\mathbb{T}_m^N$  by  $\widehat{A}_{f,m}$ . Thus,

$$(48) \quad \widehat{A}_{f,m} \in \Psi_{\text{cl}}^{-1}(\mathbb{T}_m^N; \mathbb{C}^G).$$

Since  $\kappa_m$  is a local isometry, by (47) and the transformation law for principal symbols,

$$(49) \quad \sigma_{-1}(\widehat{A}_{f,m})(\kappa_m(x), \xi) = \frac{\chi_m(x)^2}{\|\xi\|} \mathbf{c}_{f,j} \left( x, \frac{\xi}{\|\xi\|} \right), \quad x \in U_m, \quad \xi \neq 0,$$

and the principal symbol vanishes outside  $\Omega_m$ .

With respect to

$$L^2(\mathbb{T}_m^N; \mathbb{C}^G) = L^2(\Omega_m; \mathbb{C}^G) \oplus L^2(\mathbb{T}_m^N \setminus \Omega_m; \mathbb{C}^G),$$

we have

$$\widehat{A}_{f,m} = T_{f,m} \oplus 0.$$

Since  $T_{f,m}$  and  $B_{f,m}$  are unitarily equivalent,

$$(50) \quad \mu(t, \widehat{A}_{f,m}) = \mu(t, A_{f,m}), \quad t > 0.$$

By (48) and the Weyl law [?, Theorem 6.1],

$$\lim_{t \rightarrow \infty} t^{1/N} \mu(t, \widehat{A}_{f,m}) = \left( \frac{1}{N(2\pi)^N} \int_{S^* \mathbb{T}_m^N} \text{tr}_{\mathbb{C}^G} |\sigma_{-1}(\widehat{A}_{f,m})(z, \theta)|^N dz d\theta \right)^{1/N}.$$

Since the metric on  $\mathbb{T}_m^N$  is flat, combining (49) with (50), we obtain

$$(51) \quad \lim_{t \rightarrow \infty} t^{1/N} \mu(t, A_{f,m}) = \left( \frac{1}{N(2\pi)^N} \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \chi_m(x)^{2N} \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N},$$

where  $\chi_m$  is identified with its zero extension to  $\Gamma$ .

By Lemma 3.5,

$$(x, \theta) \mapsto \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N$$

belongs to  $L^1(\Gamma \times \mathbb{S}^{N-1})$ . Since  $0 \leq \chi_m \leq 1$ ,  $\chi_m \rightarrow 1$  on  $\Gamma_+$ , and  $\Gamma \setminus \Gamma_+$  is contained in finitely many hyperplanes, the dominated convergence theorem implies

$$(52) \quad \begin{aligned} & \lim_{m \rightarrow \infty} \left( \frac{1}{N(2\pi)^N} \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \chi_m(x)^{2N} \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N} \\ &= \left( \frac{1}{N(2\pi)^N} \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N}. \end{aligned}$$

By (46),  $A_{f,m} \rightarrow A_f$  in  $S^{N,\infty}$ . Combining (51), (52), and Lemma 3.8, we obtain

$$\lim_{t \rightarrow \infty} t^{1/N} \mu(t, A_f) = \left( \frac{1}{N(2\pi)^N} \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N}.$$

Since  $U$  and  $V$  are unitary,

$$\mu(t, A_f) = \mu(t, [R_{\text{Dunkl},j}, M_f]), \quad t > 0.$$

This completes the proof of Lemma 3.7.  $\square$

**Lemma 3.8.** *Let  $0 < p < \infty$ , let  $\mathcal{H}$  be a Hilbert space, and let  $A, A_m \in S^{p,\infty}(\mathcal{H})$ ,  $m \geq 1$ , satisfy*

$$\|A_m - A\|_{S^{p,\infty}} \longrightarrow 0.$$

*Suppose that, for every  $m \geq 1$ , the limit*

$$c_m := \lim_{t \rightarrow \infty} t^{1/p} \mu(t, A_m)$$

*exists. Then both limits*

$$\lim_{m \rightarrow \infty} c_m \quad \text{and} \quad \lim_{t \rightarrow \infty} t^{1/p} \mu(t, A)$$

*exist and*

$$\lim_{t \rightarrow \infty} t^{1/p} \mu(t, A) = \lim_{m \rightarrow \infty} c_m.$$

*Proof.* The assertion is the Birman–Solomyak approximation lemma; see [?, Section 11.6].  $\square$

**Lemma 3.9.** *Assume that  $N > 1$ ,  $j \in \{1, \dots, N\}$ , and  $f \in \dot{W}^{1,N}(\mathbb{R}^N)$ . Then there exists  $f_m \in C_c^\infty(\mathbb{R}^N)$  such that*

$$\begin{aligned} \|\nabla(f_m - f)\|_{L^N(\mathbb{R}^N, dx)} &\longrightarrow 0, \\ \|[R_{\text{Dunkl},j}, M_{f_m}] - [R_{\text{Dunkl},j}, M_f]\|_{S^{N,\infty}} &\longrightarrow 0, \\ \|\mathbf{c}_{f_m,j} - \mathbf{c}_{f,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))} &\longrightarrow 0. \end{aligned}$$

*Proof.* By the density of  $C_c^\infty(\mathbb{R}^N)$  in  $\dot{W}^{1,N}(\mathbb{R}^N)$ , choose  $f_m \in C_c^\infty(\mathbb{R}^N)$  such that

$$\|\nabla(f_m - f)\|_{L^N(\mathbb{R}^N, dx)} \longrightarrow 0.$$

By Theorem ?? and Lemma ??,

$$\begin{aligned} \|[R_{\text{Dunkl},j}, M_{f_m}] - [R_{\text{Dunkl},j}, M_f]\|_{S^{N,\infty}} &= \|[R_{\text{Dunkl},j}, M_{f_m-f}]\|_{S^{N,\infty}} \\ &\lesssim \|f_m - f\|_{\text{OSC}^{N,\infty}(\mathbb{R}^N)} \lesssim \|\nabla(f_m - f)\|_{L^N(\mathbb{R}^N, dx)} \longrightarrow 0. \end{aligned}$$

By the linearity of  $f \mapsto \mathbf{c}_{f,j}$  and Lemma 3.5,

$$\begin{aligned} \|\mathbf{c}_{f_m,j} - \mathbf{c}_{f,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))} &= \|\mathbf{c}_{f_m-f,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))} \\ &\lesssim_{N,R,k} \|\nabla(f_m - f)\|_{L^N(\mathbb{R}^N, dx)} \longrightarrow 0. \end{aligned}$$

This completes the proof of Lemma 3.9.  $\square$

**Theorem 3.10.** *Assume that  $N > 1$  and  $j \in \{1, \dots, N\}$ . Then, for every  $f \in \dot{W}^{1,N}(\mathbb{R}^N)$ ,*

(53)

$$\lim_{t \rightarrow \infty} t^{1/N} \mu(t, [R_{\text{Dunkl},j}, M_f]) = \left( \frac{1}{N(2\pi)^N} \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N}.$$

*Moreover,*

$$(54) \quad \left( \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N} \simeq_{N,R,k} \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}.$$

*Proof.* By Lemma 3.9, choose  $f_m \in C_c^\infty(\mathbb{R}^N)$  such that

$$(55) \quad \|[R_{\text{Dunkl},j}, M_{f_m}] - [R_{\text{Dunkl},j}, M_f]\|_{S^{N,\infty}} \longrightarrow 0,$$

$$(56) \quad \|\mathbf{c}_{f_m,j} - \mathbf{c}_{f,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))} \longrightarrow 0.$$

By Lemma 3.7, for every  $m$ ,

$$\lim_{t \rightarrow \infty} t^{1/N} \mu(t, [R_{\text{Dunkl},j}, M_{f_m}]) = \frac{1}{(N(2\pi)^N)^{1/N}} \|\mathbf{c}_{f_m,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))}.$$

By (56),

$$\|\mathbf{c}_{f_m,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))} \longrightarrow \|\mathbf{c}_{f,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))}.$$

Combining this convergence with (55), we apply Lemma 3.8 with  $p = N$  and obtain (53). Finally, (54) follows from Lemma 3.5. This completes the proof of Theorem 3.10.  $\square$

#### 4. FIRST ROCHBERG–SEMMES BASED ESTIMATE

**Theorem 4.1.** *Let  $N > 1$ . Fix a closed Weyl chamber  $\Gamma$  and  $1_G \neq g \in G$ . Suppose that  $K : \Gamma \times \Gamma \rightarrow \mathbb{C}$  satisfies*

$$|\partial_x^\alpha \partial_y^\beta K|(x, y) \leq c_{K,\alpha,\beta} \frac{|x - y|^{1-|\alpha|_1-|\beta|_1}}{|x - gy| \cdot V(x, |x - y|)}, \quad x, y \in \Gamma, \quad \alpha, \beta \in \mathbb{Z}_+^N.$$

*Let  $V$  be an integral operator with integral kernel  $K$ . For every  $f \in \dot{W}^{1,N}(\Gamma)$ , we have*

$$\|M_{f-f \circ g} V\|_{\mathcal{L}_{N,\infty}(L_2(\Gamma, w(x)dx))} \leq c_{K,\kappa} (\|f\|_{\dot{W}^{1,N}(\Gamma)} + \|f\|_{\dot{W}^{1,N}(g\Gamma)}).$$

**Theorem 4.2.** *In the setting of Theorem 4.1, let  $f \in C_c^\infty$ . If  $0 \leq \phi_m \leq 1$ ,  $m \geq 1$ , are in  $C_c^\infty(\mathbb{R}^N)$  and converge to 1 pointwise, then*

$$\text{dist}_{\mathcal{L}_{N,\infty}(L_2(\Gamma, w(x)dx))} \left( M_{\phi_m} \cdot M_f V \cdot M_{\phi_m} \rightarrow M_f V, (\mathcal{L}_{N,\infty})_0(L_2(\Gamma, w(x)dx)) \right) \rightarrow 0$$

as  $m \rightarrow \infty$ .

**Notation 4.3.** *Fix a closed Weyl chamber  $\Gamma$  and let  $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$  be the linear isomorphism appearing in Corollary 2.6.*

*Let  $\mathbb{D}_\Gamma$  be the dyadic system over  $\Gamma$  constructed in Corollary 2.7. If  $I \in \mathbb{D}_\Gamma$  and if  $Q = \Phi(I)$ , then we denote  $d_I = \ell(Q)$ .*

##### 4.1. Proof of Theorem 4.1.

**Lemma 4.4.** *There exists a smooth function  $\psi$  supported on  $[-1, 1]^N \setminus (-\frac{1}{4}, \frac{1}{4})^N$  such that*

$$1 = \sum_{I \in \mathbb{D}_\Gamma} \chi_I(x) \chi_{3I}(y) \psi\left(\frac{\Phi(x) - \Phi(y)}{d_I}\right).$$

*Proof.* Choose  $\rho \in C_c^\infty(\mathbb{R}^N)$  such that  $\rho = 1$  on  $(-\frac{1}{2}, \frac{1}{2})^N$  and  $\rho = 0$  outside  $(-1, 1)^N$ . Set  $\psi(z) = \rho(z) - \rho(2z)$ . Clearly,

$$(57) \quad \text{supp}(\psi) \subset \left\{ z \in \mathbb{R}^N : \frac{1}{4} \leq |z|_\infty \leq 1 \right\}.$$

For every  $z \neq 0$ , we have

$$(58) \quad \sum_{j \in \mathbb{Z}} \psi(2^j z) = 1.$$

For every  $x \in \Gamma$ , there exists a unique  $I \in \mathbb{D}_{\Gamma,j}$  containing  $x$ . For this  $I$ , we have  $d_I = 2^{-j}$ . We have

$$\begin{aligned} 1 &= \sum_{j \in \mathbb{Z}} \psi(2^j(\Phi(x) - \Phi(y))) = \sum_{j \in \mathbb{Z}} \left( \sum_{I \in \mathbb{D}_{\Gamma,j}} \chi_I(x) \right) \psi(2^j(\Phi(x) - \Phi(y))) = \\ &= \sum_{j \in \mathbb{Z}} \sum_{I \in \mathbb{D}_{\Gamma,j}} \chi_I(x) \psi\left(\frac{\Phi(x) - \Phi(y)}{d_I}\right) = \sum_{I \in \mathbb{D}_{\Gamma}} \chi_I(x) \psi\left(\frac{\Phi(x) - \Phi(y)}{d_I}\right). \end{aligned}$$

If  $x \in I$  and  $\psi\left(\frac{\Phi(x) - \Phi(y)}{d_I}\right) \neq 0$ , then

$$\Phi(x) \in \Phi(I) = Q, \quad |\Phi(x) - \Phi(y)| \leq d_I = \ell(Q).$$

We obviously have  $|\Phi(x) - c_Q|_{\infty} \leq \frac{1}{2}\ell(Q)$  and  $|\Phi(x) - \Phi(y)|_{\infty} \leq \ell(Q)$ . Thus,  $|\Phi(y) - c_Q| \leq \frac{3}{2}\ell(Q)$ . In other words,  $\Phi(y) \in 3Q$  or, equivalently,  $y \in 3I$ . Thus,

$$\chi_I(x) \psi\left(\frac{\Phi(x) - \Phi(y)}{d_I}\right) = \chi_I(x) \chi_{3I}(y) \psi\left(\frac{\Phi(x) - \Phi(y)}{d_I}\right).$$

This completes the proof.  $\square$

**Lemma 4.5.** Assume that  $N > 1$  and  $2 < r < \infty$ . Let  $f \in L_r^{\text{loc}}(\mathbb{R}^N)$ . Fix a closed Weyl chamber  $\Gamma$  and  $1_G \neq g \in G$ . Let  $K$  be as in Theorem 4.1. We have

$$f(x)K(x, y) = \sum_{k, l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \sum_{I \in \mathbb{D}_{\Gamma}} e_{k, l, I}(x) f_{k, l, I}(y),$$

where

- (1)  $e_{k, l, I}$  is supported in  $I$  and  $f_{k, l, I}$  is supported in  $3I$ ;
- (2) for every  $k, l \in \mathbb{Z}^N$  and for every  $I \in \mathbb{D}_{\Gamma}$ , we have

$$\|e_{k, l, I}\|_{L_r(\mathbb{R}^N, w(x)dx)} = \|f\|_{L_r(\mathbb{R}^N, w(x)dx)};$$

- (3) for every  $k, l \in \mathbb{Z}^N$  and for every  $I \in \mathbb{D}_{\Gamma}$ , we have

$$\|f_{k, l, I}\|_{L_{\infty}(\mathbb{R}^N)} \leq c_{K, \kappa, \Phi} \frac{d_I}{d_I + |c_I - gc_I|};$$

*Proof.* We have

$$\begin{aligned} K(x, y) &= \sum_{I \in \mathbb{D}_{\Gamma}} \chi_I(x) \chi_{3I}(y) K(x, y) \psi\left(\frac{\Phi(x) - \Phi(y)}{d_I}\right) = \\ &= \sum_{\substack{I \in \mathbb{D}_{\Gamma} \\ 3I \subset \Gamma}} \chi_I(x) \chi_{3I}(y) K(x, y) \psi\left(\frac{\Phi(x) - \Phi(y)}{d_I}\right) + \sum_{\substack{I \in \mathbb{D}_{\Gamma} \\ 3I \not\subset \Gamma}} \chi_I(x) \chi_{3I}(y) K(x, y) \psi\left(\frac{\Phi(x) - \Phi(y)}{d_I}\right). \end{aligned}$$

We will only consider the first sum. The second one can be treated similarly, but the notations become heavier. **feel free to write the expression for the second sum if you think it is needed. the only difference is that the domain of the map  $K_I$  becomes more complicated.**

For every  $I \in \mathbb{D}_{\Gamma}$  with  $3I \subset \Gamma$ , define the map

$$K_I : \left[-\frac{1}{2}, \frac{1}{2}\right]^N \times \left[-\frac{3}{2}, \frac{3}{2}\right]^N \rightarrow \mathbb{C}$$

by setting

$$K_I(a, b) = K(c_I + d_I \Phi^{-1}(a), c_I + d_I \Phi^{-1}(b)) \psi(a - b), \quad a \in \left[-\frac{1}{2}, \frac{1}{2}\right]^N, \quad b \in \left[-\frac{3}{2}, \frac{3}{2}\right]^N.$$

It follows from the assumption on  $K$  that **feel free to prove this if you think this requires a proof. however, do not prove it here — make a separate lemma instead.**

$$\|K_I\|_{C^{2N+2}([- \frac{1}{2}, \frac{1}{2}]^N \times [- \frac{3}{2}, \frac{3}{2}]^N)} \leq \frac{c'_{K,\kappa,\Phi}}{w(I)} \cdot \frac{d_I}{d_I + |c_I - gc_I|}.$$

Hence, there exists an extension  $K_I : \mathbb{T}^N \times \mathbb{T}^N \rightarrow \mathbb{C}$  such that

$$\|K_I\|_{C^{2N+2}(\mathbb{T}^N \times \mathbb{T}^N)} \leq \frac{c''_{K,\kappa,\Phi}}{w(I)} \cdot \frac{d_I}{d_I + |c_I - gc_I|}.$$

In particular,

$$\|K_I\|_{W^{2N+2,2}(\mathbb{T}^N \times \mathbb{T}^N)} \leq \frac{c_{K,\kappa,\Phi}}{w(I)} \cdot \frac{d_I}{d_I + |c_I - gc_I|}.$$

We may, therefore, write

$$K_I(a, b) = \sum_{k, l \in \mathbb{Z}^N} \text{coef}_{k, l, I} e^{i\langle k, a \rangle} e^{i\langle l, b \rangle}, \quad a, b \in \mathbb{T}^N,$$

where

$$\sum_{k, l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{2N+2} |\text{coef}_{k, l, I}|^2 \leq \left( \frac{c_{K,\kappa,\Phi}}{w(I)} \cdot \frac{d_I}{d_I + |c_I - gc_I|} \right)^2.$$

In particular,

$$(1 + |k|^2 + |l|^2)^{-N-1} \cdot |\text{coef}_{k, l, I}| \leq \frac{c_{K,\kappa,\Phi}}{w(I)} \cdot \frac{d_I}{d_I + |c_I - gc_I|}, \quad k, l \in \mathbb{Z}^N, \quad I \in \mathbb{D}_\Gamma.$$

We now set

$$e_{k, l, I}(x) = f(x) \chi_I(x) e^{i\langle k, \frac{\Phi^{-1}(x - c_I)}{d_I} \rangle},$$

$$f_{k, l, I}(y) = (1 + |k|^2 + |l|^2)^{N+1} \text{coef}_{k, l, I} \cdot \chi_{3I}(y) e^{i\langle l, \frac{\Phi^{-1}(y - c_I)}{d_I} \rangle}.$$

This allows us to write

$$f(x) K(x, y) = \sum_{k, l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \sum_{I \in \mathbb{D}_\Gamma} e_{k, l, I}(x) f_{k, l, I}(y).$$

Obviously,

$$\text{supp}(e_{k, l, I}) \subset I, \quad \text{supp}(f_{k, l, I}) \subset 3I,$$

$$\|e_{k, l, I}\|_{L_r(\mathbb{R}^N, w(x)dx)} = \|f\|_{L_r(I, w(x)dx)},$$

$$\|f_{k, l, I}\|_{L_\infty(\mathbb{R}^N, w(x)dx)} = (1 + |k|^2 + |l|^2)^{N+1} |\text{coef}_{I, k, l}| \leq \frac{c_{K,\kappa,\Phi}}{w(I)} \cdot \frac{d_I}{d_I + |c_I - gc_I|}.$$

□

**Lemma 4.6.** *For every  $I \in \mathbb{D}_\Gamma$ , we have*

$$\sup_{x \in I} w(x) \leq c_{\kappa, \Phi} \frac{w(I)}{m(I)}.$$

*Proof.* We have

$$w(I) \approx_{\Phi, \kappa} w(B(c_I, d_I)) \approx_{\kappa} d_I^N \prod_{v \in R} (|\langle c_I, v \rangle| + d_I)^{k(v)}.$$

On the other hand, if  $x \in I$ , then  $|x - c_I| \leq c_\Phi d_I$  and, therefore,

$$|\langle x, v \rangle| \leq |\langle x - c_I, v \rangle| + |\langle c_I, v \rangle| \leq |x - c_I| + |\langle c_I, v \rangle| \leq c_\Phi d_I + |\langle c_I, v \rangle|.$$

Thus,

$$w(x) \leq \prod_{v \in R} (c_\Phi d_I + |\langle c_I, v \rangle|)^{k(v)} \leq c_{\Phi, \kappa} \prod_{v \in R} (|\langle c_I, v \rangle| + d_I)^{k(v)}.$$

This completes the proof.  $\square$

**Lemma 4.7.** *Assume that  $N > 1$  and  $2 < r < \infty$ . Let  $f \in L_r^{\text{loc}}(\mathbb{R}^N)$ . We have*

$$\|M_f V\|_{\mathcal{L}_{N, \infty}(L_2(\Gamma, w(x)dx))} \leq c_{K, \kappa, \Phi, r} \left\| \left\{ m(I)^{-\frac{1}{r}} \|f\|_{L_r(I)} \cdot \frac{d_I}{d_I + |c_I - g c_I|} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{N, \infty}.$$

*Proof.* Let  $A_{k, l}$  be the integral operator with integral kernel given by the formula

$$(x, y) \rightarrow \sum_{I \in \mathbb{D}_\Gamma} e_{k, l, I}(x) f_{k, l, I}(y), \quad x, y \in \Gamma, \quad k, l \in \mathbb{Z}^N.$$

By Lemma 4.5, we have

$$M_f V = \sum_{k, l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} A_{k, l}.$$

By the triangle inequality,

$$\begin{aligned} & \|M_f V\|_{\mathcal{L}_{N, \infty}(L_2(\Gamma, w(x)dx))} \leq \\ & \leq \frac{N}{N-1} \sum_{k, l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \|A_{k, l}\|_{\mathcal{L}_{N, \infty}(L_2(\Gamma, w(x)dx))}. \end{aligned}$$

By Lemma 4.5, the sequences

$$\left\{ w(I)^{\frac{1}{r} - \frac{1}{2}} \frac{e_{k, l, I}}{\|f\|_{L_r(I, w(x)dx)}} \right\}_{I \in \mathbb{D}_\Gamma}, \quad \left\{ c_{K, \kappa, \Phi}^{-1} w(I)^{\frac{1}{2}} \cdot \frac{d_I + |c_I - g c_I|}{d_I} f_{k, l, I} \right\}_{I \in \mathbb{D}_\Gamma}$$

are  $(r, 1)$ -NWO and, respectively,  $(\infty, 3)$ -NWO in the sense of Definition 1.6. Using Lemma 1.7, we conclude that

$$\begin{aligned} & \|A_{k, l}\|_{\mathcal{L}_{N, \infty}(L_2(\Gamma, w(x)dx))} \leq \\ & \leq c_{\Phi, \kappa, r, 1, \infty, 3, N} c_{K, \kappa} \left\| \left\{ w(I)^{-\frac{1}{r}} \|f\|_{L_r(I, w(x)dx)} \cdot \frac{d_I}{d_I + |c_I - g c_I|} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{N, \infty}. \end{aligned}$$

Therefore,

$$\begin{aligned} & \|M_f V\|_{\mathcal{L}_{N, \infty}(L_2(\Gamma, w(x)dx))} \leq \\ & \leq \frac{N}{N-1} c_{\Phi, \kappa, r, 1, \infty, 3, N} c_{K, \kappa, \Phi} \sum_{k, l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \times \\ & \times \left\| \left\{ w(I)^{-\frac{1}{r}} \|f\|_{L_r(I, w(x)dx)} \cdot \frac{d_I}{d_I + |c_I - g c_I|} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{N, \infty}. \end{aligned}$$

By Lemma 4.6,

$$w(I)^{-\frac{1}{r}} \|f\|_{L_r(I, w(x)dx)} \leq c_{\kappa, \Phi}^{\frac{1}{r}} m(I)^{-\frac{1}{r}} \|f\|_{L_r(I)}, \quad I \in \mathbb{D}_\Gamma.$$

Combining the last two estimates, we complete the proof.  $\square$

**Lemma 4.8.**

$$\left\| \left\{ m(I)^{-1} \|f \circ g\|_{L_1(I)} \cdot \frac{d_I}{d_I + |c_I - g c_I|} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{N, \infty} \leq c_\Phi (\|f\|_{\dot{W}^{1, N}(\Gamma)} + \|f\|_{\dot{W}^{1, N}(g\Gamma)}).$$

*Proof.* Let  $\mathcal{M}$  be the Hardy–Littlewood maximal operator on  $\mathbb{R}^N$ . By [?, Theorem 3], for almost every  $x \in \mathbb{R}^N$ ,

$$|f(x) - f(gx)| \leq c_N |x - gx| \cdot ((\mathcal{M}(|\nabla f|))(x) + (\mathcal{M}(|\nabla f|))(gx)).$$

We have

$$|x - gx| \leq 2|x - c_I| + |c_I - gc_I|, \quad x \in I, \quad y \in gI.$$

Obviously,

$$|x - c_I| \leq \frac{d_I^{\frac{1}{2}}}{2} \|\Phi\|_\infty d_I, \quad x \in I.$$

Hence,

$$|x - gx| \leq \max\{1, d_I^{\frac{1}{2}} \|\Phi\|_\infty\} \cdot (d_I + |c_I - gc_I|), \quad x \in I, \quad y \in gI.$$

Thus,

$$|f(x) - f(gx)| \leq c'_\Phi \cdot (d_I + |c_I - gc_I|) \cdot ((\mathcal{M}(|\nabla f|))(x) + (\mathcal{M}(|\nabla f|))(gx)).$$

This immediately implies

$$\frac{1}{d_I + |c_I - gc_I|} \|f - f \circ g\|_{L_1(I)} \leq c''_\Phi \cdot (\|\mathcal{M}(|\nabla f|)\|_{L_1(I)} + \|\mathcal{M}(|\nabla f|)\|_{L_1(gI)})$$

for every  $I \in \mathbb{D}_\Gamma$ . Hence,

$$\begin{aligned} m(I)^{-1} \|f - f \circ g\|_{L_1(I)} \cdot \frac{d_I}{d_I + |c_I - gc_I|} &\leq \\ &\leq c'''_\Phi \cdot m(I)^{\frac{1}{N}-1} (\|\mathcal{M}(|\nabla f|)\|_{L_1(I)} + \|\mathcal{M}(|\nabla f|)\|_{L_1(gI)}) \end{aligned}$$

for every  $I \in \mathbb{D}_\Gamma$ . Thus,

$$\begin{aligned} &\left\| \left\{ m(I)^{-1} \|f - f \circ g\|_{L_1(I)} \cdot \frac{d_I}{d_I + |c_I - gc_I|} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{N, \infty} \leq \\ &\leq c'''_\Phi \frac{N}{N-1} \left( \left\| \left\{ m(I)^{\frac{1}{N}-1} \|\mathcal{M}(|\nabla f|)\|_{L_1(I)} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{N, \infty} + \right. \\ &\quad \left. + \left\| \left\{ m(I)^{\frac{1}{N}-1} \|\mathcal{M}(|\nabla f|)\|_{L_1(gI)} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{N, \infty} \right) \leq \\ &\stackrel{L.5.1}{\leq} c'''_\Phi \frac{N}{N-1} c_N (\|\mathcal{M}(|\nabla f|)\|_{L_N(\Gamma)} + \|\mathcal{M}(|\nabla f|)\|_{L_N(g\Gamma)}) \leq \\ &\leq c'''_\Phi \frac{N}{N-1} c_N \|\mathcal{M}\|_{L_N(\mathbb{R}^N) \odot} (\|\nabla f\|_{L_N(\Gamma)} + \|\nabla f\|_{L_N(g\Gamma)}). \end{aligned}$$

□

**Lemma 4.9.** *Let  $2 < r < N$  and let  $f \in \dot{W}^{1,N}(\Gamma) \cap \dot{W}^{1,N}(g\Gamma)$ . We have*

$$\left\| \left\{ m(I)^{-\frac{1}{r}} \|f - f \circ g\|_{L_r(I)} \cdot \frac{d_I}{d_I + |c_I - gc_I|} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{N, \infty} \leq c_\Phi (\|f\|_{\dot{W}^{1,N}(\Gamma)} + \|f\|_{\dot{W}^{1,N}(g\Gamma)}).$$

*Proof.* We obviously have

$$\begin{aligned} \|f - f \circ g\|_{L_r(I)} &\leq \left\| f - \frac{1}{m(I)} \int_I f \right\|_{L_r(I)} + \\ &+ \left\| f \circ g - \frac{1}{m(I)} \int_I (f \circ g) \right\|_{L_r(I)} + m(I)^{\frac{1}{r}-1} \|f - f \circ g\|_{L_1(I)}. \end{aligned}$$

By Poincare inequality,

$$\|f - f \circ g\|_{L_r(I)} \leq$$

$$\leq c_{\Phi,r} \left( m(I)^{\frac{1}{N}} \|\nabla f\|_{L_r(I)} + m(I)^{\frac{1}{N}} \|\nabla(f \circ g)\|_{L_r(I)} + m(I)^{\frac{1}{r}-1} \|f - f \circ g\|_{L_1(I)} \right).$$

By the triangle inequality,

$$\begin{aligned} & \left\| \left\{ m(I)^{-\frac{1}{r}} \|f - f \circ g\|_{L_r(I)} \cdot \frac{d_I}{d_I + |c_I - gc_I|} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{N,\infty} \leq \\ & \leq \frac{N}{N-1} c_{\Phi,r} \left\| \left\{ m(I)^{\frac{1}{N}-\frac{1}{r}} \|\nabla f\|_{L_r(I)} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{N,\infty} + \\ & + \frac{N}{N-1} c_{\Phi,r} \left\| \left\{ m(I)^{\frac{1}{N}-\frac{1}{r}} \|\nabla f\|_{L_r(gI)} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{N,\infty} + \\ & + \frac{N}{N-1} c_{\Phi,r} \left\| \left\{ m(I)^{-1} \|f - f \circ g\|_{L_1(I)} \cdot \frac{d_I}{d_I + |c_I - gc_I|} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{N,\infty}. \end{aligned}$$

The assertion follows now from Lemma 5.1 (for the first two summands) and Lemma 4.8 (for the last summand).  $\square$

*Proof of Theorem 4.1.* The assertion follows by combining Lemmas 4.7 (taken with  $f - f \circ g$  instead of  $f$ ) and 4.9.  $\square$

#### 4.2. Proof of Theorem 4.2.

**Lemma 4.10.** *Let  $f, \phi \in C_c^\infty(\mathbb{R}^N)$ . If  $0 \leq \phi \leq 1$ , then*

$$\begin{aligned} & \text{dist}_{\mathcal{L}_{N,\infty}(L_2(\Gamma, w(x)dx))} \left( M_f V - M_{f\phi} V, (\mathcal{L}_{N,\infty})_0(L_2(\Gamma, w(x)dx)) \right) \leq c_{K,\kappa,\phi,r} \times \\ & \times \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left( \left\{ \frac{d_I m(I)^{-\frac{1}{r}}}{d_I + |c_I - gc_I|} \|f\|_{L_r(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right), \\ & \text{dist}_{\mathcal{L}_{N,\infty}(L_2(\Gamma, w(x)dx))} \left( M_f V - M_f V M_\phi, (\mathcal{L}_{N,\infty})_0(L_2(\Gamma, w(x)dx)) \right) \leq c_{K,\kappa,\phi,r} \times \\ & \times \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left( \left\{ \frac{d_I m(I)^{-\frac{1}{r}}}{d_I + |c_I - gc_I|} \|f\|_{L_r(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right). \end{aligned}$$

*Proof.* We only prove the second inequality. The proof of the first one is similar, but simpler.

Let  $A_{k,l}$  be as in the proof of Lemma 4.7. We have

$$M_f V - M_f V M_\phi = \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} A_{k,l} M_{1-\phi}.$$

By the triangle inequality,

$$\begin{aligned} & \text{dist}_{\mathcal{L}_{N,\infty}(L_2(\Gamma, w(x)dx))} \left( M_f V - M_f V M_\phi, (\mathcal{L}_{N,\infty})_0(L_2(\Gamma, w(x)dx)) \right) \leq \\ & \leq \frac{N}{N-1} \sum_{k,l \in \mathbb{Z}^N} \left( (1 + |k|^2 + |l|^2)^{-N-1} \times \right. \\ & \left. \times \text{dist}_{\mathcal{L}_{N,\infty}(L_2(\Gamma, w(x)dx))} \left( A_{k,l} M_{1-\phi}, (\mathcal{L}_{N,\infty})_0(L_2(\Gamma, w(x)dx)) \right) \right). \end{aligned}$$

By Lemma 4.5, the sequences

$$\left\{ w(I)^{\frac{1}{r}-\frac{1}{2}} \frac{e_{k,l,I}}{\|f\|_{L_r(I, w(x)dx)}} \right\}_{I \in \mathbb{D}_\Gamma}, \quad \left\{ c_{K,\kappa,\Phi}^{-1} w(I)^{\frac{1}{2}} \cdot \frac{d_I + |c_I - gc_I|}{d_I} f_{k,l,I} \cdot (1-\phi) \right\}_{I \in \mathbb{D}_\Gamma}$$

are  $(r, 1)$ -NWO and, respectively,  $(\infty, 3)$ -NWO in the sense of Definition 1.6. Obviously,  $e_{k,l,I} \neq 0$  only if  $I \cap \text{supp}(f) \neq \emptyset$  and  $f_{k,l,I} \cdot (1 - \phi) \neq 0$  only if  $I \cap \text{supp}(f) \neq \emptyset$  and  $3I \cap \text{supp}(1 - \phi) \neq \emptyset$ . Using Lemma 1.7, we conclude that

$$\begin{aligned} & \text{dist}_{\mathcal{L}_{N,\infty}(L_2(\Gamma, w(x)dx))} \left( A_{k,l}(1 - \phi), (\mathcal{L}_{N,\infty})_0(L_2(\Gamma, w(x)dx)) \right) \leq \\ & \leq c_{\Phi, \kappa, r, 1, \infty, 3, N} c_{K, \kappa, \Phi} \times \\ & \times \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left( \left\{ \frac{d_I w(I)^{-\frac{1}{r}}}{d_I + |c_I - gc_I|} \|f\|_{L_r(I, w(x)dx)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1 - \phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right). \end{aligned}$$

By Lemma 4.6,

$$w(I)^{-\frac{1}{r}} \|f\|_{L_r(I, w(x)dx)} \leq c_{\kappa, \Phi}^{\frac{1}{r}} m(I)^{-\frac{1}{r}} \|f\|_{L_r(I)}, \quad I \in \mathbb{D}_\Gamma.$$

Thus,

$$\begin{aligned} & \text{dist}_{\mathcal{L}_{N,\infty}(L_2(\Gamma, w(x)dx))} \left( A_{k,l}(1 - \phi), (\mathcal{L}_{N,\infty})_0(L_2(\Gamma, w(x)dx)) \right) \leq \\ & \leq c_{\Phi, \kappa, r, 1, \infty, 3, N} c_{K, \kappa, \Phi} c_{\kappa, \Phi}^{\frac{1}{r}} \times \\ & \times \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left( \left\{ \frac{d_I m(I)^{-\frac{1}{r}}}{d_I + |c_I - gc_I|} \|f\|_{L_r(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1 - \phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right). \end{aligned}$$

Combining this estimate with the one in the first paragraph **assign label and refer to it** yields the second inequality.  $\square$

**Lemma 4.11.** *Let  $2 < r < N$ . For every  $f \in C_c^\infty(\mathbb{R}^N)$  and for every  $n \in \mathbb{N}$ , we have*

$$\begin{aligned} & \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left( \left\{ m(I)^{\frac{1}{N} - \frac{1}{r}} \|\nabla f\|_{L_r(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1 - \phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right) \leq \\ & \leq \|\nabla f\|_{L_\infty(\mathbb{R}^N)} \times \left\| \left\{ m(I)^{\frac{1}{N}} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1 - \phi) \neq \emptyset \\ d_I \leq 2^{-n}}} \right\|_{l_{N,\infty}(\mathbb{D}_\Gamma)}. \end{aligned}$$

*Proof.* For a given  $n \in \mathbb{Z}$ , we have

$$\left\{ m(I)^{\frac{1}{N} - \frac{1}{r}} \|\nabla f\|_{L_r(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1 - \phi) \neq \emptyset \\ d_I > 2^{-n}}} \leq \|\nabla f\|_{L_r(\mathbb{R}^N)} \left\{ m(I)^{\frac{1}{N} - \frac{1}{r}} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1 - \phi) \neq \emptyset \\ d_I > 2^{-n}}}.$$

The right hand side obviously belongs to  $(l_{N,\infty})_0(\mathbb{D}_\Gamma)$  and, hence, so does the left hand side. Thus,

$$\begin{aligned} & \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left( \left\{ m(I)^{\frac{1}{N} - \frac{1}{r}} \|\nabla f\|_{L_r(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1 - \phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right) \leq \\ & \leq \left\| \left\{ m(I)^{\frac{1}{N} - \frac{1}{r}} \|\nabla f\|_{L_r(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1 - \phi) \neq \emptyset \\ d_I \leq 2^{-n}}} \right\|_{l_{N,\infty}(\mathbb{D}_\Gamma)} \leq \\ & \leq \|\nabla f\|_{L_\infty(\mathbb{R}^N)} \times \left\| \left\{ m(I)^{\frac{1}{N}} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1 - \phi) \neq \emptyset \\ d_I \leq 2^{-n}}} \right\|_{l_{N,\infty}(\mathbb{D}_\Gamma)}. \end{aligned}$$

□

**Lemma 4.12.** *For every  $f \in C_c^\infty(\mathbb{R}^N)$  and for every  $n \in \mathbb{N}$ , we have*

$$\begin{aligned} \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left( \left\{ \frac{d_I m(I)^{-1}}{d_I + |c_I - gc_I|} \|f - f \circ g\|_{L_1(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right) \leq \\ \leq c_\Phi \|\nabla f\|_{L_\infty(\mathbb{R}^N)} \times \left\| \left\{ m(I)^{\frac{1}{N}} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset \\ d_I \leq 2^{-n}}} \right\|_{l_{N,\infty}(\mathbb{D}_\Gamma)}. \end{aligned}$$

*Proof.* For a given  $n \in \mathbb{Z}$ , we have

$$\left\{ \frac{d_I m(I)^{-1}}{d_I + |c_I - gc_I|} \|f - f \circ g\|_{L_1(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset \\ d_I > 2^{-n}}} \leq 2 \|f\|_{L_1(\mathbb{R}^N)} \left\{ m(I)^{-1} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}.$$

The right hand side obviously belongs to  $(l_{N,\infty})_0(\mathbb{D}_\Gamma)$  and, hence, so does the left hand side. Thus,

$$\begin{aligned} \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left( \left\{ \frac{d_I m(I)^{-1}}{d_I + |c_I - gc_I|} \|f - f \circ g\|_{L_1(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right) \leq \\ \leq \left\| \left\{ \frac{d_I m(I)^{-1}}{d_I + |c_I - gc_I|} \|f - f \circ g\|_{L_1(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset \\ d_I \leq 2^{-n}}} \right\|_{l_{N,\infty}(\mathbb{D}_\Gamma)}. \end{aligned}$$

Obviously,

$$|f(x) - f(y)| \leq c_N |x - y| \cdot \|\nabla f\|_{L_\infty(\mathbb{R}^N)}.$$

Hence,

$$\begin{aligned} \|f - f \circ g\|_{L_1(I)} &\leq c_N \|\nabla f\|_{L_\infty(\mathbb{R}^N)} \times \int_I |x - gx| dx \leq \\ &\leq c'_\Phi m(I) \times (d_I + |c_I - gc_I|) \times \|\nabla f\|_{L_\infty(\mathbb{R}^N)}. \end{aligned}$$

Thus,

$$\frac{d_I m(I)^{-1}}{d_I + |c_I - gc_I|} \|f - f \circ g\|_{L_1(I)} \leq c_\Phi m(I)^{\frac{1}{N}} \|\nabla f\|_{L_\infty(\mathbb{R}^N)}$$

Finally,

$$\begin{aligned} \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left( \left\{ \frac{d_I m(I)^{-1}}{d_I + |c_I - gc_I|} \|f - f \circ g\|_{L_1(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right) \leq \\ \leq c_\Phi \|\nabla f\|_{L_\infty(\mathbb{R}^N)} \times \left\| \left\{ m(I)^{\frac{1}{N}} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset \\ d_I \leq 2^{-n}}} \right\|_{l_{N,\infty}(\mathbb{D}_\Gamma)}. \end{aligned}$$

□

**Lemma 4.13.** *For every  $f \in C_c^\infty(\mathbb{R}^N)$  and for every  $n \in \mathbb{N}$ , we have*

$$\begin{aligned} \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left( \left\{ \frac{d_I m(I)^{-\frac{1}{r}}}{d_I + |c_I - g c_I|} \|f - f \circ g\|_{L_r(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right) \leq \\ \leq c_\Phi \|\nabla f\|_{L_\infty(\mathbb{R}^N)} \times \left\| \left\{ m(I)^{\frac{1}{N}} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset \\ d_I \leq 2^{-n}}} \right\|_{l_{N,\infty}(\mathbb{D}_\Gamma)}. \end{aligned}$$

*Proof.* We obviously have

$$\begin{aligned} \|f - f \circ g\|_{L_r(I)} &\leq \left\| f - \frac{1}{m(I)} \int_I f \right\|_{L_r(I)} + \\ &+ \left\| f \circ g - \frac{1}{m(I)} \int_I (f \circ g) \right\|_{L_r(I)} + m(I)^{\frac{1}{r}-1} \|f - f \circ g\|_{L_1(I)}. \end{aligned}$$

By Poincare inequality,

$$\begin{aligned} \|f - f \circ g\|_{L_r(I)} &\leq \\ &\leq c_{\Phi,r} \left( m(I)^{\frac{1}{N}} \|\nabla f\|_{L_r(I)} + m(I)^{\frac{1}{N}} \|\nabla(f \circ g)\|_{L_r(I)} + m(I)^{\frac{1}{r}-1} \|f - f \circ g\|_{L_1(I)} \right). \end{aligned}$$

By the triangle inequality,

$$\begin{aligned} \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left( \left\{ \frac{d_I m(I)^{-\frac{1}{r}}}{d_I + |c_I - g c_I|} \|f - f \circ g\|_{L_r(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right) \leq \\ \leq \frac{N}{N-1} c_{\Phi,r} \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left( \left\{ m(I)^{\frac{1}{N}-\frac{1}{r}} \|\nabla f\|_{L_r(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right) + \\ + \frac{N}{N-1} c_{\Phi,r} \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left( \left\{ m(I)^{\frac{1}{N}-\frac{1}{r}} \|\nabla f\|_{L_r(gI)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right) + \\ + \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left( \left\{ \frac{d_I m(I)^{-1}}{d_I + |c_I - g c_I|} \|f - f \circ g\|_{L_1(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right). \end{aligned}$$

□

## 5. APPENDIX

**Lemma 5.1.** *Let  $1 < s < \infty$ , and let  $\mathbb{D}^0$  be the standard dyadic grid in  $\mathbb{R}^N$ . Suppose that  $F \in L^s(\mathbb{R}^N)$ . Then there exists a constant  $C > 0$  such that*

$$\left\| \left\{ |Q|^{-1/s'} \int_Q |F(x)| dx \right\}_{Q \in \mathbb{D}^0} \right\|_{\ell^{s,\infty}} \leq C \|F\|_{L^s(\mathbb{R}^N)}.$$

*Proof.* Let  $\mathcal{E} \subset \mathbb{D}^0$  be finite and set

$$G_{\mathcal{E}}(x) := \sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \chi_Q(x).$$

Since the dyadic cubes containing any fixed  $x$  are totally ordered by inclusion, for each  $x \in \bigcup_{Q \in \mathcal{E}} Q$  we may choose a cube  $Q_x \in \mathcal{E}$  of minimal volume such that

$x \in Q_x$ . Every  $Q \in \mathcal{E}$  containing  $x$  is then a dyadic ancestor of  $Q_x$ . Since the volume increases by the factor  $2^N$  at each generation,

$$G_{\mathcal{E}}(x) = \sum_{\substack{Q \in \mathcal{E} \\ x \in Q}} |Q|^{-1/s'} \leq |Q_x|^{-1/s'} \sum_{k=0}^{\infty} 2^{-kN/s'} \lesssim_{N,s} \sup_{Q \in \mathcal{E}} |Q|^{-1/s'} \chi_Q(x).$$

Consequently,

$$\|G_{\mathcal{E}}\|_{L^{s'}(\mathbb{R}^N)}^{s'} \lesssim_{N,s} \int_{\mathbb{R}^N} \sup_{Q \in \mathcal{E}} |Q|^{-1} \chi_Q(x) dx \leq \int_{\mathbb{R}^N} \sum_{Q \in \mathcal{E}} |Q|^{-1} \chi_Q(x) dx = \#\mathcal{E}.$$

and hence,

$$\|G_{\mathcal{E}}\|_{L^{s'}(\mathbb{R}^N)} \lesssim_{N,s} (\#\mathcal{E})^{1/s'}.$$

By Hölder's inequality,

$$\sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \int_Q |F(x)| dx = \int_{\mathbb{R}^N} |F(x)| G_{\mathcal{E}}(x) dx \lesssim_{N,s} \|F\|_{L^s(\mathbb{R}^N)} (\#\mathcal{E})^{1/s'}.$$

Fix  $t > 0$  and let  $\mathcal{E}$  be any finite subcollection of the dyadic cubes satisfying

$$|Q|^{-1/s'} \int_Q |F(x)| dx > t.$$

Then

$$t \#\mathcal{E} < \sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \int_Q |F(x)| dx \lesssim_{N,s} \|F\|_{L^s(\mathbb{R}^N)} (\#\mathcal{E})^{1/s'},$$

so

$$\#\mathcal{E} \lesssim_{N,s} t^{-s} \|F\|_{L^s(\mathbb{R}^N)}^s.$$

Since  $\mathcal{E}$  is arbitrary,

$$\#\left\{Q \in \mathbb{D}^0 : |Q|^{-1/s'} \int_Q |F(x)| dx > t\right\} \lesssim_{N,s} t^{-s} \|F\|_{L^s(\mathbb{R}^N)}^s, \quad t > 0.$$

The asserted weak  $\ell^s$  estimate follows. This completes the proof of Lemma 5.1.  $\square$

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